

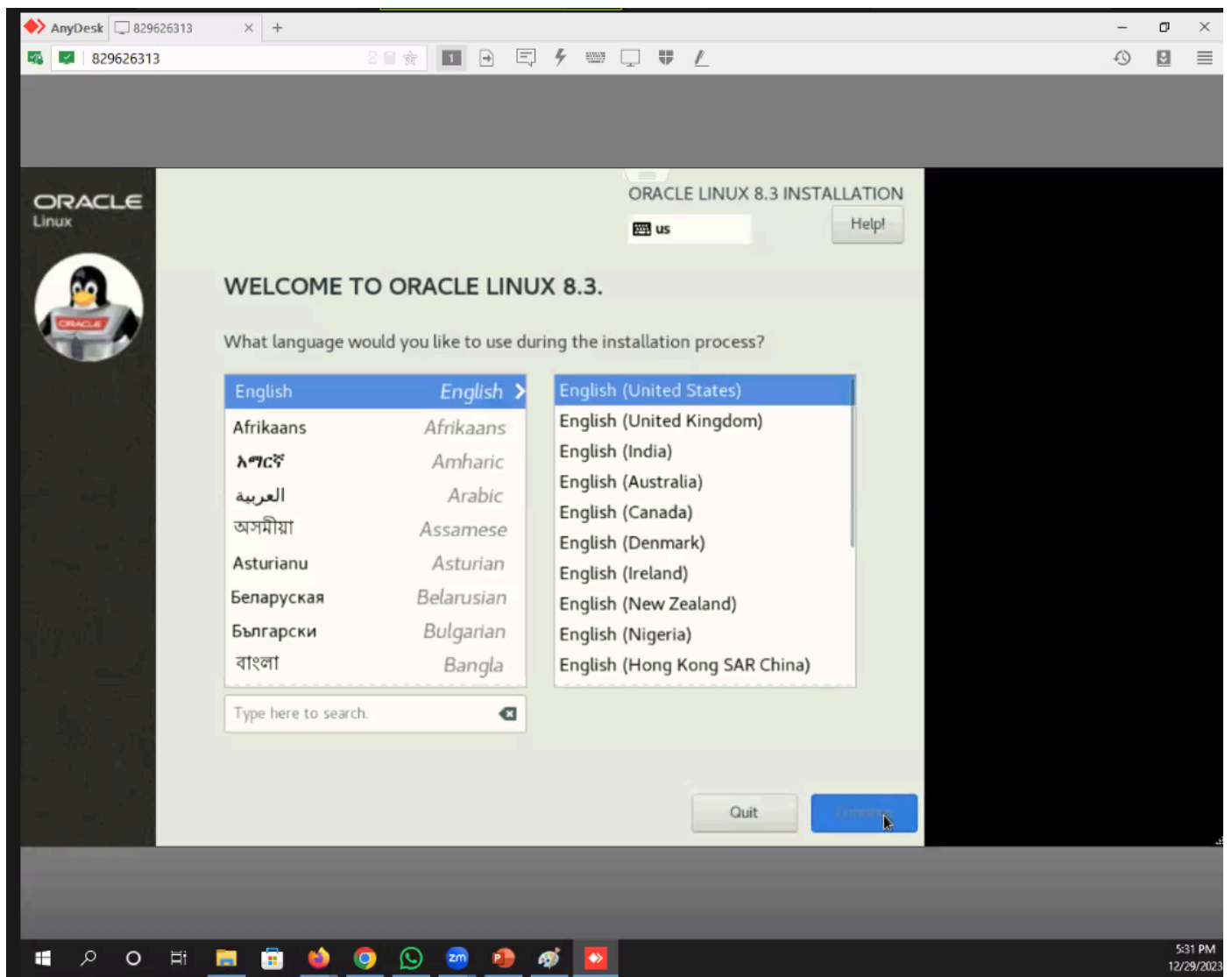
1.oracle DBA Linux & oracle 19c installation

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we will be working on oracle 19c

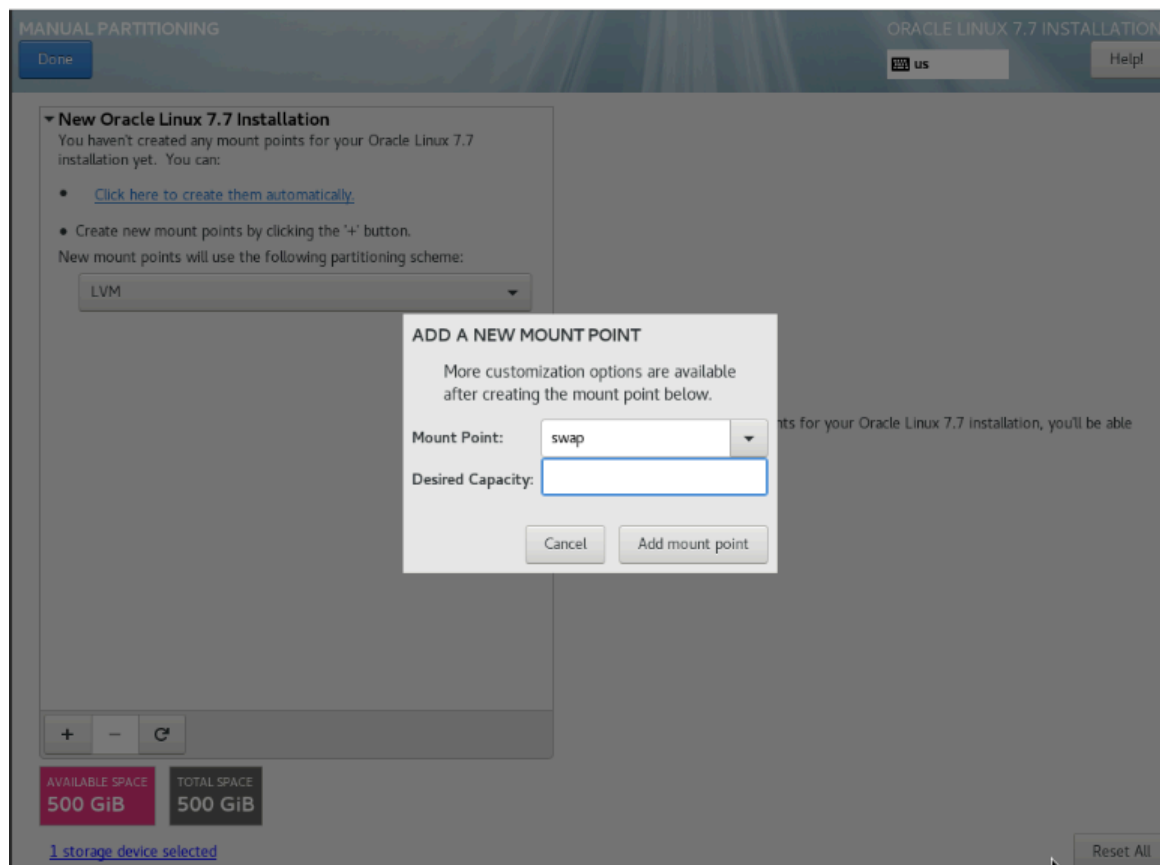
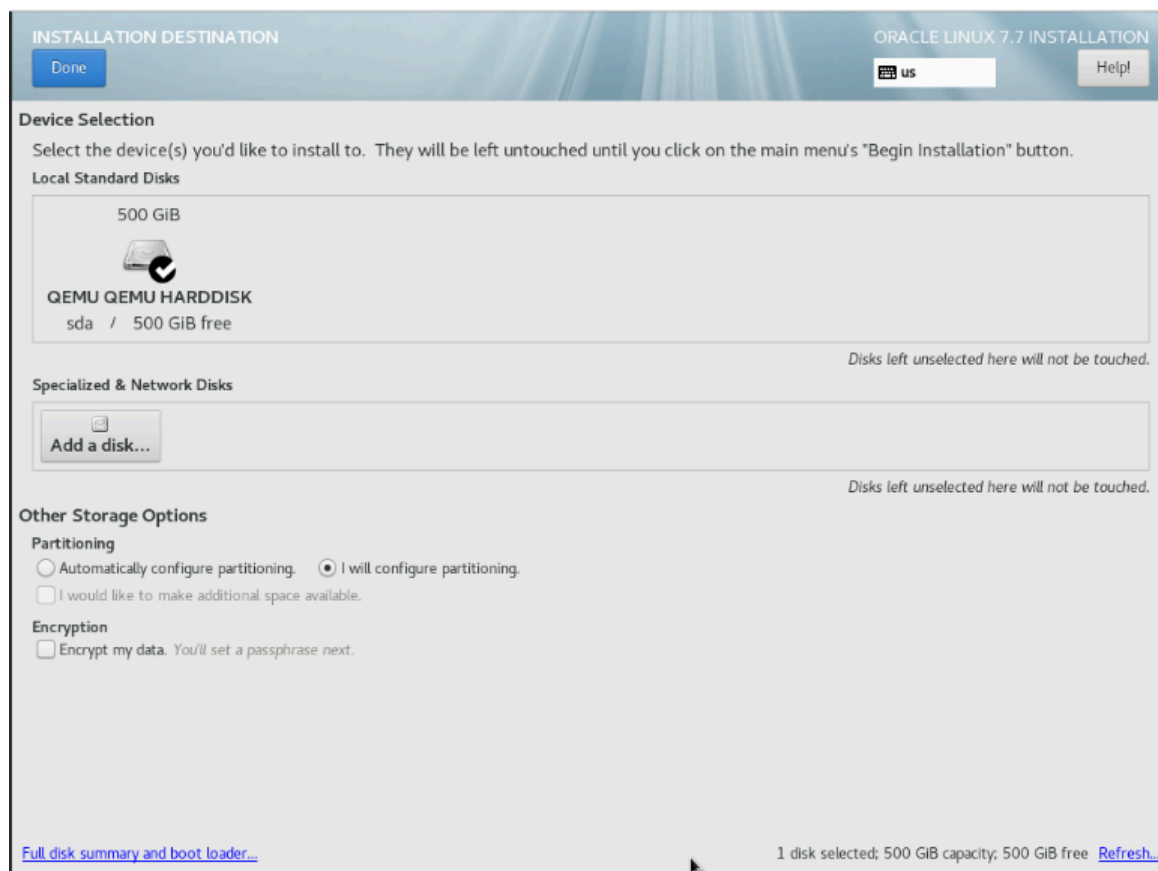
it will be installed on oracle Linux 7.

installation oracle Linux 7



[redhat instalation](#)

- you will have to setup partition .
- disable kdm feature
- select data and time
- setup root user
- setup network
- setup hostname



after installation prerequisites

first you need update and upgrade .

```
yum update -y
yum upgrade -y
```

xmllrpc-c-client	x86_64	1.51.0-8.el8	ol8_baseos_latest	213 k
xmllrpc-c-client	x86_64	1.51.0-8.el8	ol8_baseos_latest	41 k
xorg-x11-dev-intel	x86_64	2.99.917-41.20210115.el8	ol8_appstream	706 k
xorg-x11-font-utils	x86_64	1:7.5-41.el8	ol8_appstream	184 k
xorg-x11-server-Xorg	x86_64	1.20.11-17.el8	ol8_appstream	1.5 M
xorg-x11-server-Xwayland	x86_64	21.1.3-12.el8	ol8_appstream	966 k
xorg-x11-server-common	x86_64	1.20.11-17.el8	ol8_appstream	44 k
xz	x86_64	5.2.4-4.el8_6	ol8_baseos_latest	153 k
xz-libs	x86_64	5.2.4-4.el8_6	ol8_baseos_latest	94 k
yaql	x86_64	2.1.0-12.el8	ol8_appstream	41 k
yum	noarch	4.7.8-19.0.1.el8	ol8_baseos_latest	289 k
zenity	x86_64	3.28.1-2.el8	ol8_appstream	4.0 M
zlib	x86_64	1.2.11-25.el8	ol8_baseos_latest	182 k
Installing dependencies:				
NetworkManager-initscripts-updown	noarch	1:1.40.16-0.0.1.el8	ol8_baseos_latest	143 k
autogen-libopts	x86_64	5.18.12-8.el8	ol8_appstream	75 k
clang-resource-filesystem	x86_64	16.0.6-2.0.1.module+el8.9.0+90022+019b57b8	ol8_appstream	15 k
crda	noarch	3.18.2020.04.29-1.el8	ol8_baseos_latest	24 k
environment-modules	x86_64	4.5.2-4.el8	ol8_baseos_latest	422 k
fstrm	x86_64	0.6.1-3.el8	ol8_appstream	29 k
fuse3	x86_64	3.3.0-17.0.1.el8	ol8_baseos_latest	54 k
gcc-toolset-13-binutils	x86_64	2.40-14.0.1.el8	ol8_appstream	6.9 M
gcc-toolset-13-binutils-gold	x86_64	2.40-14.0.1.el8	ol8_appstream	947 k
gcc-toolset-13-gcc	x86_64	13.1.1-4.3.el8	ol8_appstream	48 M
gcc-toolset-13-gcc-c++	x86_64	13.1.1-4.3.el8	ol8_appstream	14 M
gcc-toolset-13-libstdc++-devel	x86_64	13.1.1-4.3.el8	ol8_appstream	3.6 M
gcc-toolset-13-runtime	x86_64	13.0-2.el8	ol8_appstream	28 k
glibc-gconv-extra	x86_64	2.28-236.0.1.el8_7	ol8_baseos_latest	1.6 M
gnutls-dane	x86_64	3.6.16-7.el8	ol8_appstream	52 k
gnutls-utils	x86_64	3.6.16-7.el8	ol8_appstream	349 k
grub2-tools-efi	x86_64	1:2.02-150.0.1.el8	ol8_baseos_latest	481 k
iw	x86_64	5.19-1.el8_1	ol8_baseos_latest	120 k
kernel2x-firmware	noarch	999:20231102-999.29.el8	ol8_baseos_latest	25 M
keyutils-libs-devel	x86_64	1.5.10-9.el8	ol8_baseos_latest	48 k
krb5-devel	x86_64	1.18.2-26.0.1.el8_9	ol8_baseos_latest	561 k
libcom_err-devel	x86_64	1.45.6-5.el8	ol8_baseos_latest	39 k
libfdt	x86_64	1.6.0-1.el8	ol8_appstream	32 k
libkadm5	x86_64	1.18.2-26.0.1.el8_9	ol8_baseos_latest	187 k
libselinux-devel	x86_64	2.9-8.el8	ol8_baseos_latest	288 k
libsepol-devel	x86_64	2.9-3.el8	ol8_baseos_latest	87 k
librpm	x86_64	0.9.1-2.20211126git1ff6f6f43.module+el8.9.0+90052+d3bf71d8	ol8_appstream	184 k
libtraceevent	x86_64	1.5.3-1.el8	ol8_baseos_latest	217 k
libvirt-devel	x86_64	0.3.2-2.el8	ol8_baseos_latest	18 k
libwpe	x86_64	1.10.0-4.el8	ol8_appstream	26 k
libzstd-devel	x86_64	1.4.4.1.0.1.el8	ol8_baseos_latest	44 k
linux-firmware-core	noarch	999:20231102-999.29.git2b304bfe.el8	ol8_baseos_latest	2.1 M
lmdb-libs	x86_64	0.9.24-2.el8	ol8_baseos_latest	58 k

installing oracle DB 19c

[Install Database Only.txt](#)

[Error.txt](#)

step 1, update host file

you need to go your host file and add ip and hostname of the current server

you can use the below command that will that will indent the hostname to the file for faster action.

```
hostname >> /etc/hosts
```

the syntax is for updating hostname is as follow

```
ip hostname FQDN
```

then vi and add the Ip address

oracle-db01

```
:::1      localhost localhost.localdomain localhost6 localhost6.localdomain6
10.10.10.26 orcaledb      orcaledb.local
```


step 2, install required packages

there are some prerequisite packages that need to be installed you can copy whole script and paste it in the terminal

```
yum install -y bc
yum install -y binutils
yum install -y compat-libcap1
yum install -y compat-libstdc++-33
yum install -y dtrace-modules
yum install -y dtrace-modules-headers
yum install -y dtrace-modules-provider-headers
yum install -y dtrace-utils
yum install -y elfutils-libelf
yum install -y elfutils-libelf-devel
yum install -y fontconfig-devel
yum install -y glibc
yum install -y glibc-devel
yum install -y ksh
yum install -y libaio
yum install -y libaio-devel
yum install -y libdtrace-ctf-devel
yum install -y libXrender
yum install -y libXrender-devel
yum install -y libX11
yum install -y libXau
yum install -y libXi
yum install -y libXtst
yum install -y libgcc
yum install -y librdmacm-devel
yum install -y libstdc++
yum install -y libstdc++-devel
yum install -y libxcb
yum install -y make
yum install -y net-tools
yum install -y nfs-utils
yum install -y python
yum install -y python-configshell
yum install -y python-rtslib
yum install -y python-six
yum install -y targetcli
yum install -y smartmontools
yum install -y sysstat
```

you can also use the below command that will install the required package but will not consider only the default data path and user, which limits our ability to modify the installation.

```
yum install -y oracle-database-preinstall-19c
```

it will install the default path and default settings so that's why we are using the manual way to install packages.

step 3 create the groups

the below command will create groups that are needed for successful installation

```
groupadd -g 54321 oinstall
groupadd -g 54322 dba
groupadd -g 54323 oper
groupadd -g 54324 backupdba
groupadd -g 54325 dgdba
groupadd -g 54326 kmdba
```

```
groupadd -g 54327 asmdba
groupadd -g 54328 asmoper
groupadd -g 54329 asmadmin
groupadd -g 54330 racdba
```

Complete!

```
[root@orcaledb ~]# groupadd -g 54321 oinstall
[root@orcaledb ~]# groupadd -g 54322 dba
[root@orcaledb ~]# groupadd -g 54323 oper
[root@orcaledb ~]# groupadd -g 54324 backupdba
[root@orcaledb ~]# groupadd -g 54325 dgdba
[root@orcaledb ~]# groupadd -g 54326 kmdba
[root@orcaledb ~]# groupadd -g 54327 asmdba
[root@orcaledb ~]# groupadd -g 54328 asmoper
[root@orcaledb ~]# groupadd -g 54329 asmadmin
[root@orcaledb ~]# groupadd -g 54330 racdba
[root@orcaledb ~]# groupadd -g 54321 oinstall
groupadd: group 'oinstall' already exists
[root@orcaledb ~]# groupadd -g 54322 dba
groupadd: group 'dba' already exists
[root@orcaledb ~]# groupadd -g 54323 oper
groupadd: group 'oper' already exists
[root@orcaledb ~]# groupadd -g 54324 backupdba
groupadd: group 'backupdba' already exists
[root@orcaledb ~]# groupadd -g 54325 dgdba
groupadd: group 'dgdba' already exists
[root@orcaledb ~]# groupadd -g 54326 kmdba
groupadd: group 'kmdba' already exists
[root@orcaledb ~]# groupadd -g 54327 asmdba
groupadd: group 'asmdba' already exists
[root@orcaledb ~]# groupadd -g 54328 asmoper
groupadd: group 'asmoper' already exists
[root@orcaledb ~]# groupadd -g 54329 asmadmin
groupadd: group 'asmadmin' already exists
[root@orcaledb ~]# groupadd -g 54330 racdba
groupadd: group 'racdba' already exists
[root@orcaledb ~]# █
```

I have run the command again to make sure that group are done

step 4 , create the users and add them to the groups

we need to create user called oracle that will be used to install and managed oracle database

the root will be only used to managed the server but oracle is used to managed db

below command will create users and add them to group `oinstall` , and also it will create a user called oracle that will be used you can change the name of the user if you would like

`oinstall` group is responsible for installation of oracle db

you can notice there is other users such as dba , backupdba that have there part in daily operation of oracle DB

```
useradd -u 54321 -g oinstall -G dba,oper,backupdba,dgdba,kmdba,asmdba,asmoper,asmadmin,racdba oracle
```

steps 5, update selinux config

To allow Oracle to be installed and make changes on the OS level, you need to change the permission on SELinux. You can do this by following the steps below:

- Open the file `/etc/selinux/config` using the vi editor.
- Add the following line to the file: `SELINUX=permissive`.
- Save the file and exit the editor.

This should allow Oracle to be installed and make changes on the OS level.

```
As User Oracle
```

```
nano /etc/selinux/config
```

```
set SELINUX=permissive
```

```
Last login: Sat Dec 30 13:00:42 2023 from 10.10.10.4
[oracle@orcaledb ~]$ less /etc/selinux/config
# This file controls the state of SELinux on the system.
SELINUX=permissive can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
SELINUX=enforcing
# SELINUXTYPE= can take one of three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted

/etc/selinux/config (END)
```

```
selinux
```

```
setenforce Permissive
```



```
mkdir /u01/app/oracle/product/19c/db_1
mkdir -p /u01/oradata
chown -R oracle:oinstall /u01
chmod -R 775 /u01

mkdir -p /u01/app/oracle/product/12.2.1
chown -R oracle:oinstall /u01/app/oracle/product/12.2.1
chmod -R 775 /u01/app/oracle/product/12.2.1
```

step 8 edit the bash profile of oracle user

next you need to edit the bash profile to specify the directory of oracle data and bin library

the directory must match the the same name directories in server and in `export ORACLE_HOSTNAME=` MAKE SURE YOU FILL THE SERVER Hostname

```
nano /home/oracle/.bash_profile

# Oracle Settings
export TMP=/tmp
export TMPDIR=$TMP

export ORACLE_HOSTNAME=db
export ORACLE_UNQNAME=orcl
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=$ORACLE_BASE/product/19c/db_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=orcl
export PDB_NAME=pdorcl
export DATA_DIR=/u01/oradata
export PATH=/usr/sbin:/usr/local/bin:$PATH
export PATH=$ORACLE_HOME/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib
```

```
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH
export TMP=/tmp
export TMPDIR=$TMP

export ORACLE_HOSTNAME=orcaledb.local
export ORACLE_UNQNAME=orcl
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=$ORACLE_BASE/product/19c/db_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=orcl
export PDB_NAME=pdorcl
export DATA_DIR=/u01/oradata
export PATH=/usr/sbin:/usr/local/bin:$PATH
export PATH=$ORACLE_HOME/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib

:wg!
```

step 9 upload oracle db. installation

upload the database file and move the file to **using the** oracle user****

then copy the file to the below directory we have created before

note : above procces must be done using oracle user and not the root

```
/u01/app/oracle/product/19c/db_1/
```

then unzip the file make sure all this steps must be run using the oracle user

run the installer

```
./runInstaller
```

note : When dealing with Oracle installation on a server running a non-GUI (Graphical User Interface) operating system, a different approach is needed. In this scenario, the absence of a GUI means you can't directly interact with the installer through graphical means.

To overcome this limitation, you'll have to initiate an SSH (Secure Shell) connection from a separate device that does have a GUI and is equipped with Java JDK (Java Development Kit). This secondary device will serve as your remote control center for the installation process.

step 10 resolve installation error

sometimes the installation it wont work due error and you will need to add some directory to make installation GUI open on source device , in this case for me I will SSH to the server through windows server running mobxtrem that have java already installed on it , next you need to update some libraries on home directory , you will add the below line that will allow source device java to display the installation gui, below must be done using root user

```
nano /home/oracle/.bash profile
```

```
DISPLAY=[device that ssh to the server]:0.0; export DISPLAY
```

```

# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH
export TMP=/tmp
export TMPDIR=$TMP

export ORACLE_HOSTNAME=orcaledb.local
export ORACLE_UNQNAME=orcl
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=$ORACLE_BASE/product/19c/db_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=orcl
export PDB_NAME=pdorcl
export DATA_DIR=/u01/oradata
export PATH=/usr/sbin:/usr/local/bin:$PATH
export PATH=$ORACLE_HOME/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib
DISPLAY=10.10.10.25:0.0; export DISPLAY
~
~
~
~
~
~
~
"/home/oracle/.bash profile" 28L, 727C

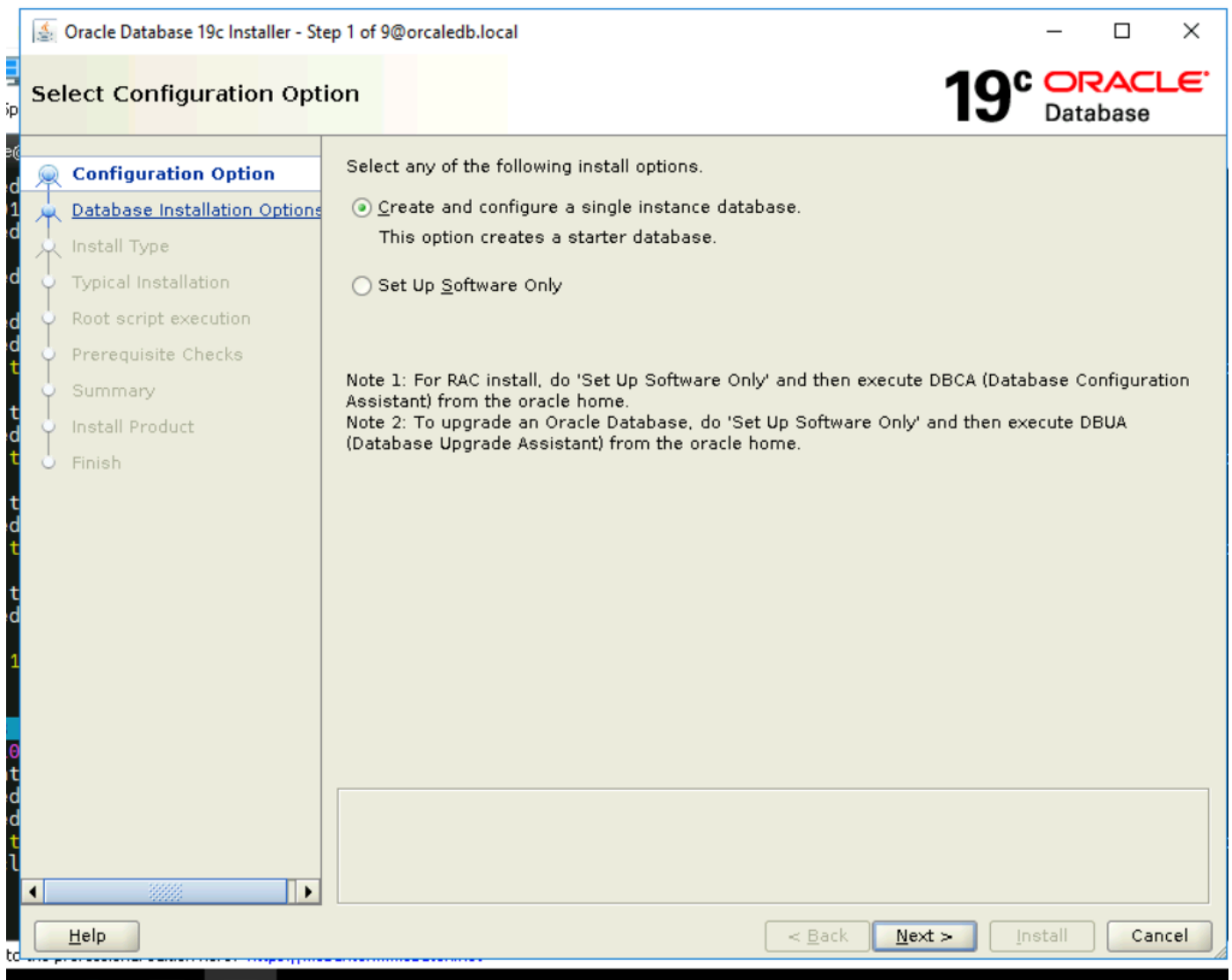
```

[Error.txt](#)

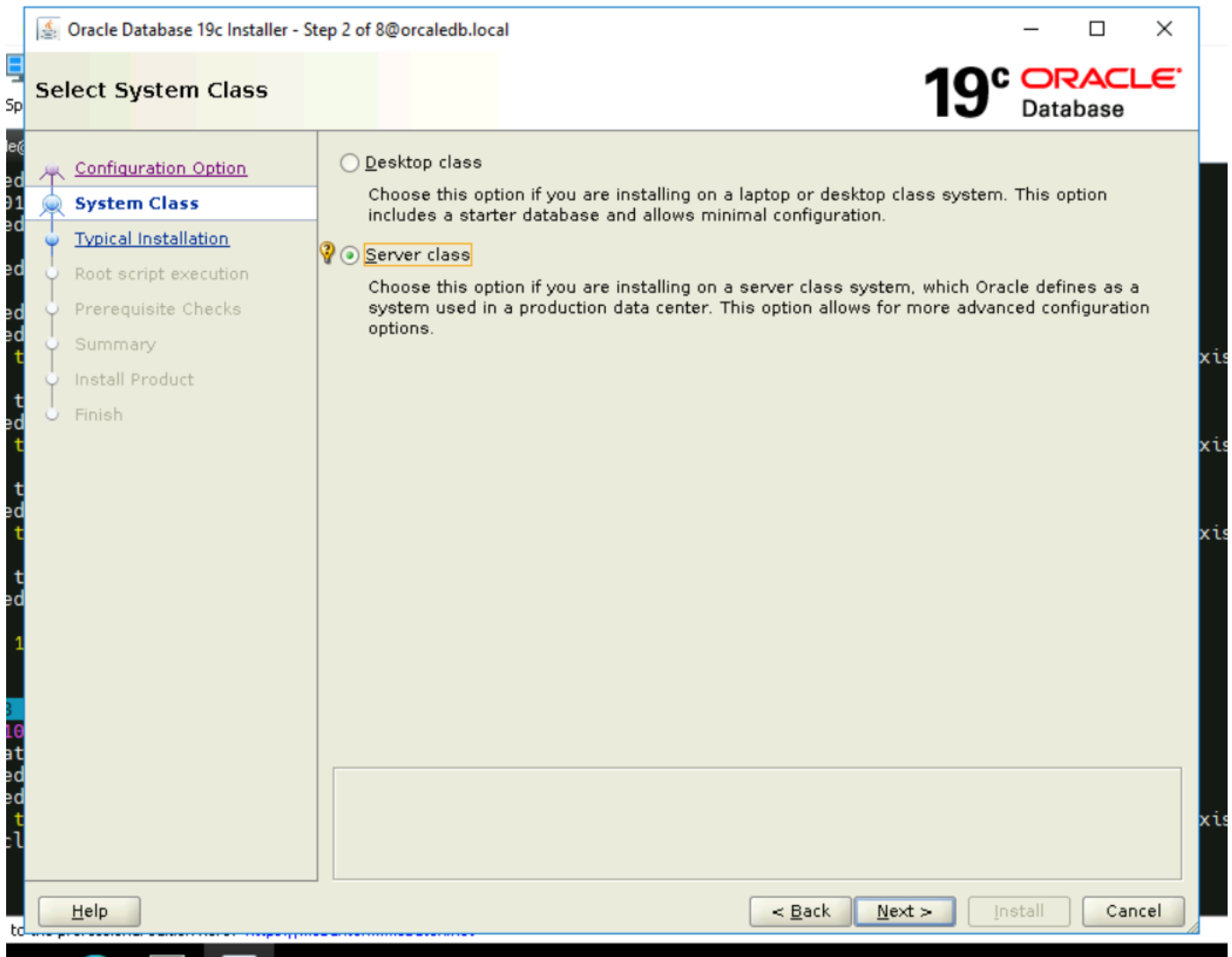
steps 11 complete the installation wizard

with wizard started you will be asked to select between two option

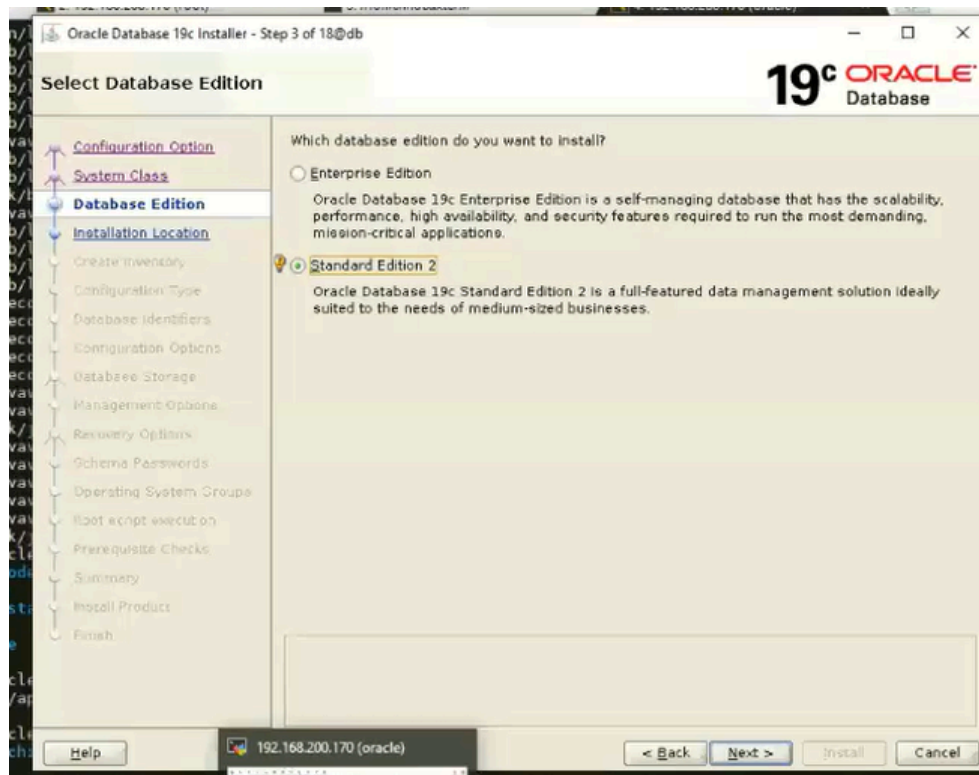
- 1.create and configure instance: this the option we will select to install and configure instance
 2. setup software only , this will only install the database engine useful for RAC implementation
- select 'create and configure a single instance database '



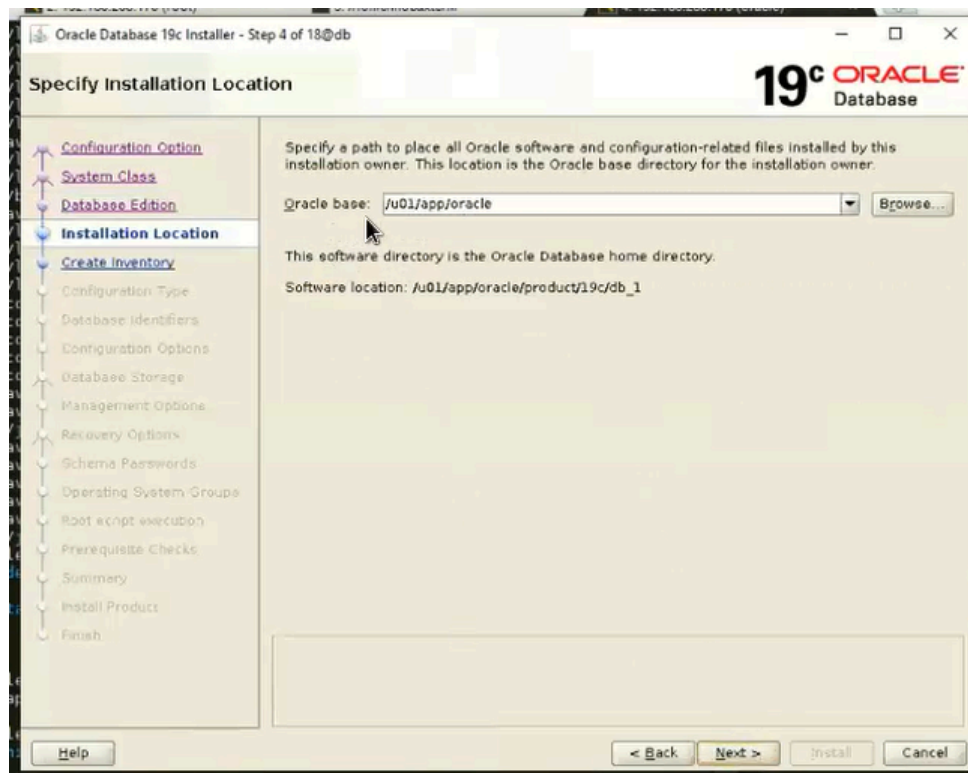
1. **desktop class** : will have minimal configuration useful if you want to install oracle on you personal laptop
2. **server class** : contain all the feature an has advanced configuration suitable for production
we will select **server class**



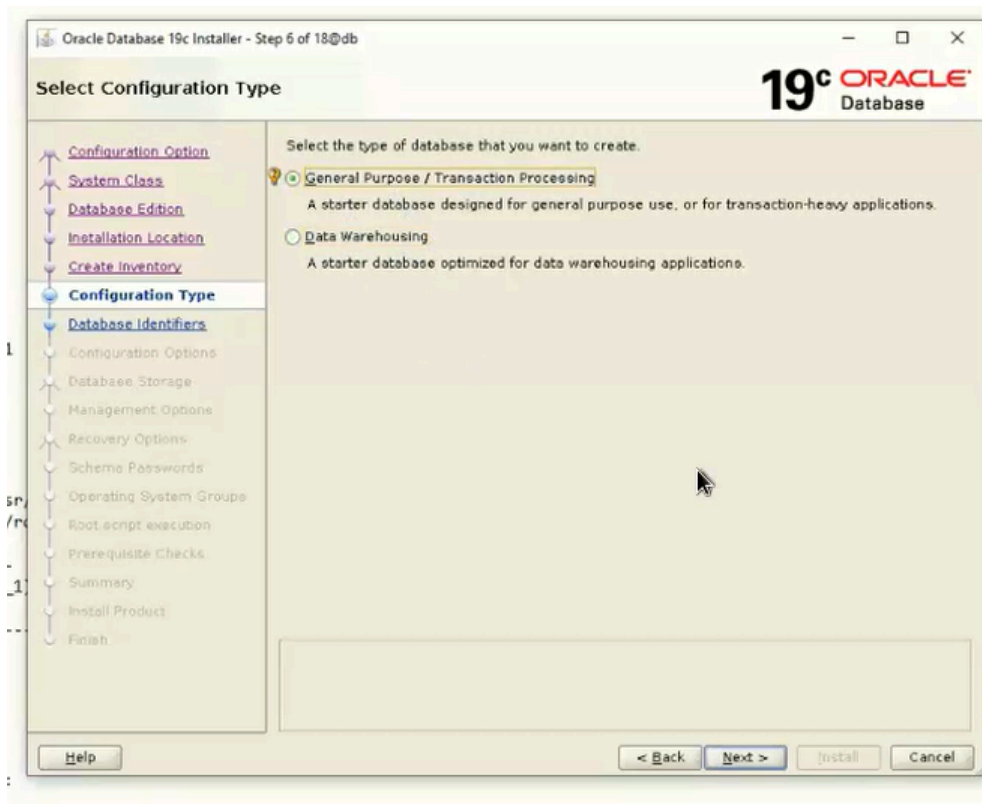
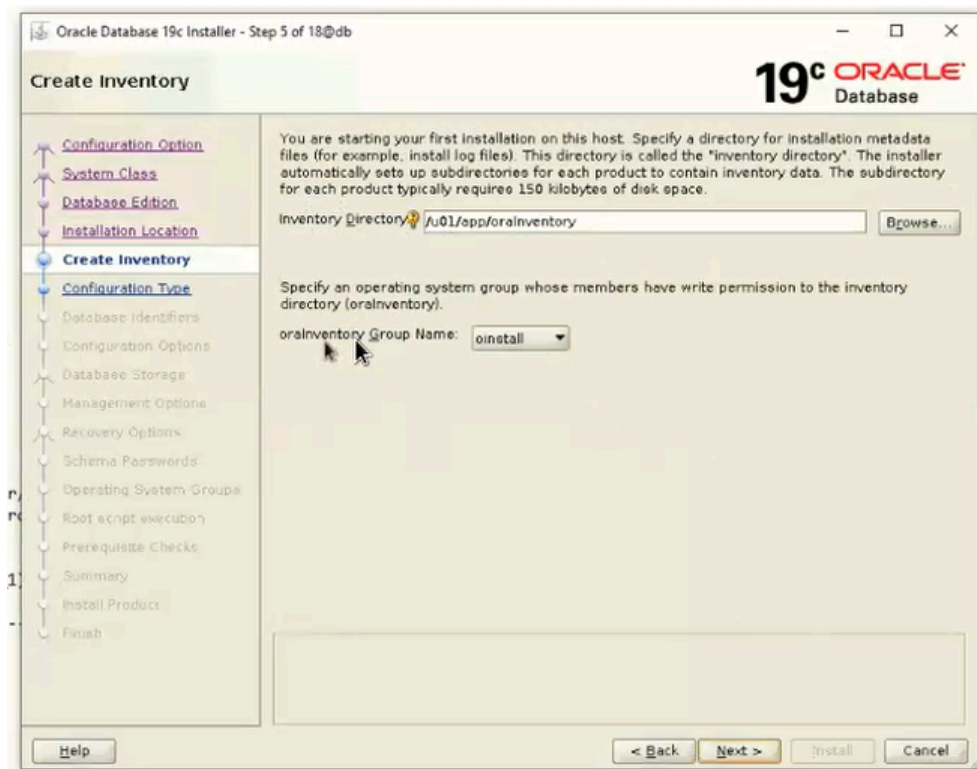
1. **standard** : allow only to setup one database container
database container is collection of database allow similar to PostgreSQL dbcluster
2. **enterprise** : allow to create as much as db container as you want also it support high availability and more security feature
for this purpose we will select enterprise edition



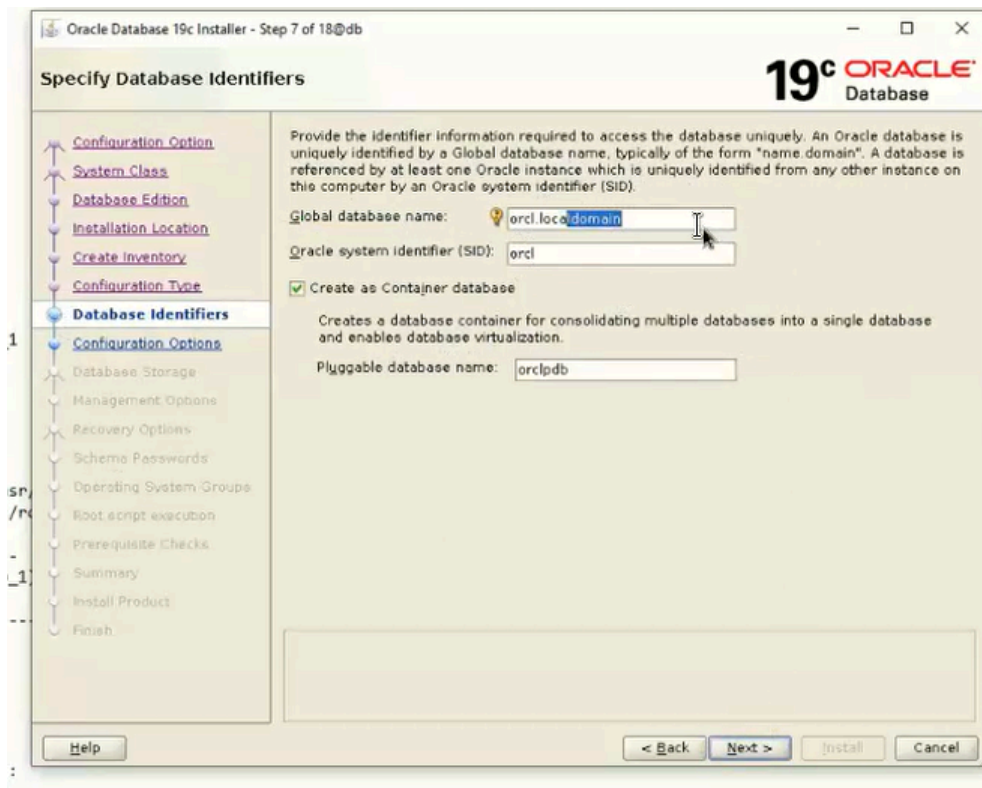
next window will ask to confirm oracle base path just confirm its the same in the server and bash profile and click next .



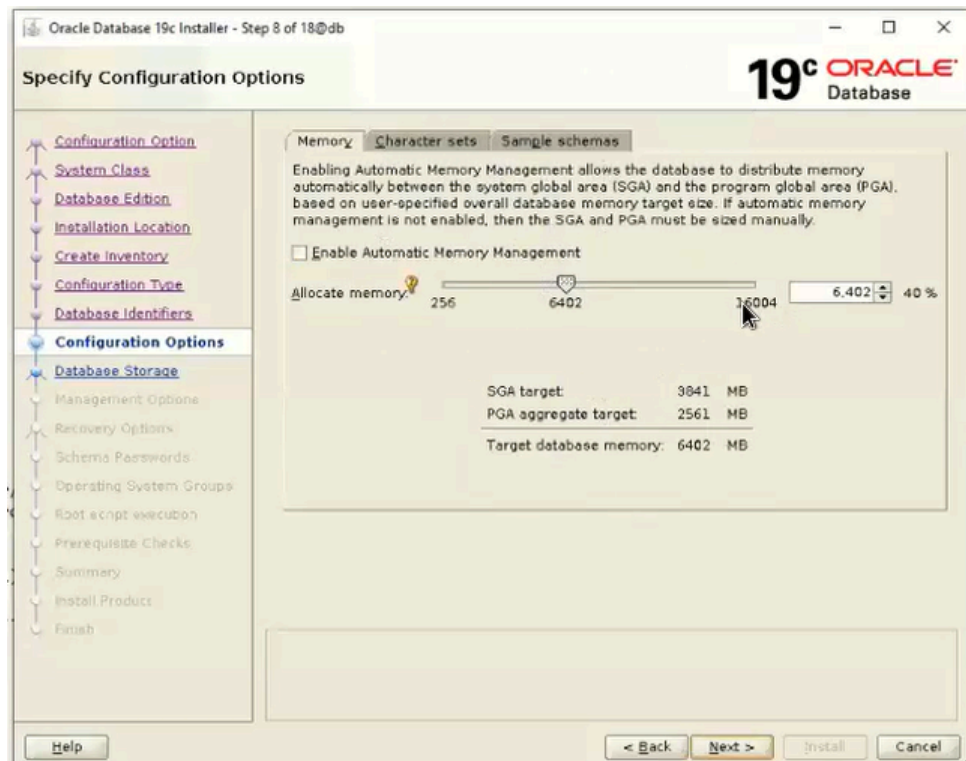
next window also you have to confirm inventory directory to be same in the server , also confirm the group 'oinstall' must be available on sever



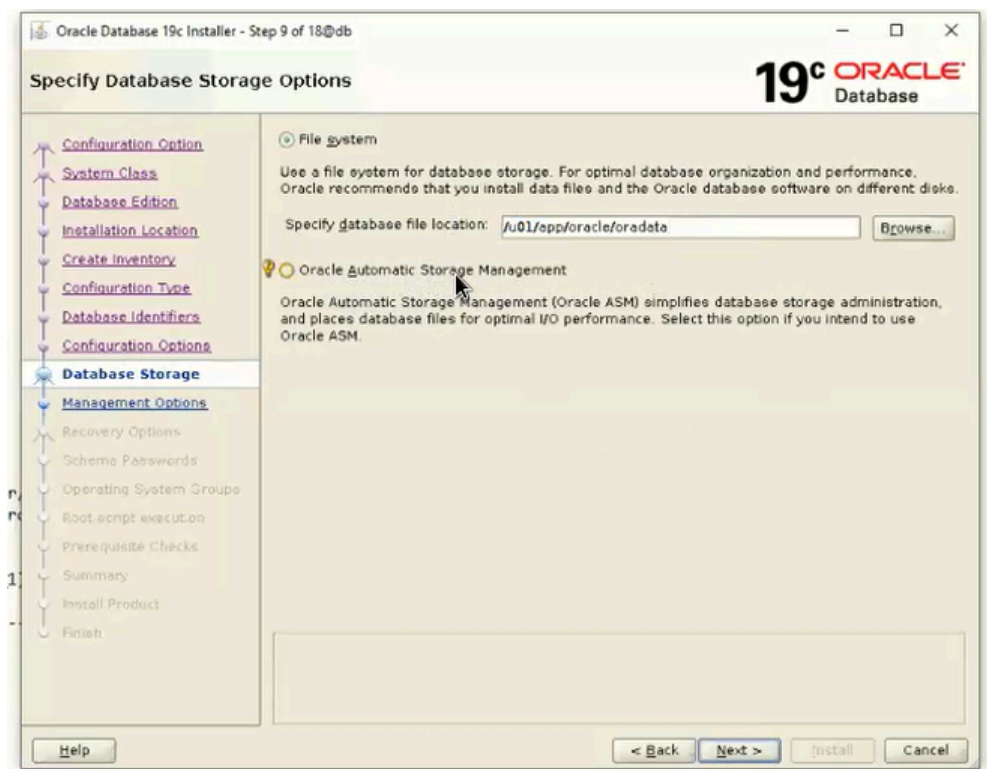
next window will ask about global database , just make
sure they match the bash profile



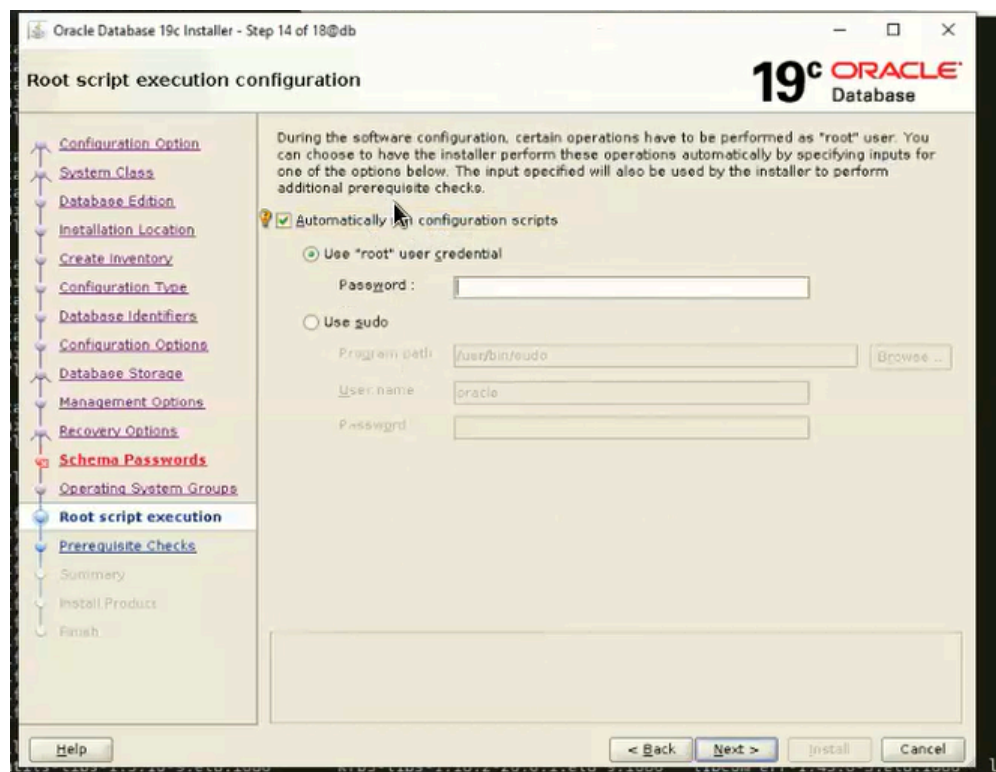
next window about memory memory usage you can restrict it or keep the default , also there character set similar to ms sql collation , simple schema it will ask to add sample schema to test database

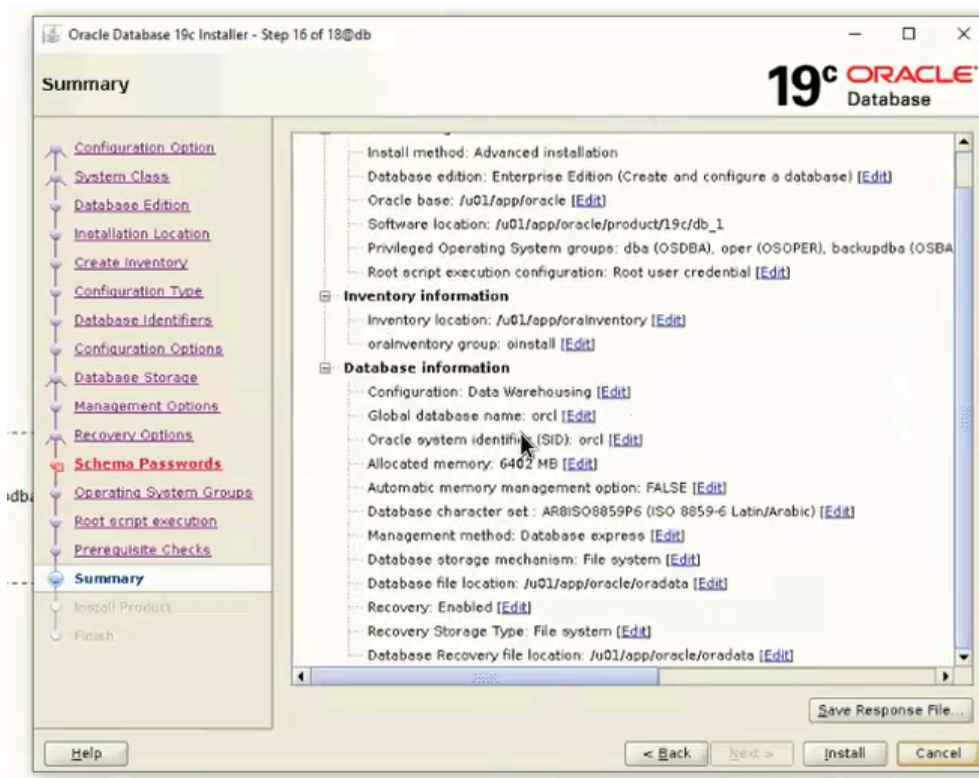


next window will ask where to store data , just confirm and click next



next window root script execution , there some script that need privileged user to execute the window will ask to provide root username and password or a user that have sedure privilege , uncheck mark automatly run he configuration and click next means there possible chance the installation will failed , check mark it and provide either root or suder username and password





connecting to oracle dba

to connect to oracle database make sure your running using oracle & use the below command

```
sqlplus sys/sys as sysdba
```

```
[oracle@orcaledb db_1]$ sqlplus sys/sys as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Dec 30 12:51:49 2023
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> █
```

use the below command to show which user is running

```
show user
```

starting database after restart

in case of the server crash or you restart the server

oracle database will not start automatically


```
SQL*Plus: Release 19.0.0.0.0 - Production on Sat Dec 30 13:01:59 2023
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to an idle instance.

SQL>
```

first run the below command to start the listener

```
lsnrctl start
```

```
[oracle@orcaledb ~]$ lsnrctl start

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 30-DEC-2023 13:03:15

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting /u01/app/oracle/product/19c/db_1/bin/tnslsnr: please wait...

TNSLSNR for Linux: Version 19.0.0.0.0 - Production
System parameter file is /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Log messages written to /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=1521)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                LISTENER
Version              TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date           30-DEC-2023 13:03:17
Uptime                0 days 0 hr. 0 min. 0 sec
Trace Level           off
Security              ON: Local OS Authentication
SNMP                  OFF
Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File     /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))
The listener supports no services
The command completed successfully
[oracle@orcaledb ~]$
```

```
startup
```

once done the "database open "must show means the database is online

```

SQL> stratup
SP2-0042: unknown command "stratup" - rest of line ignored.
SQL> startup
ORACLE instance started.

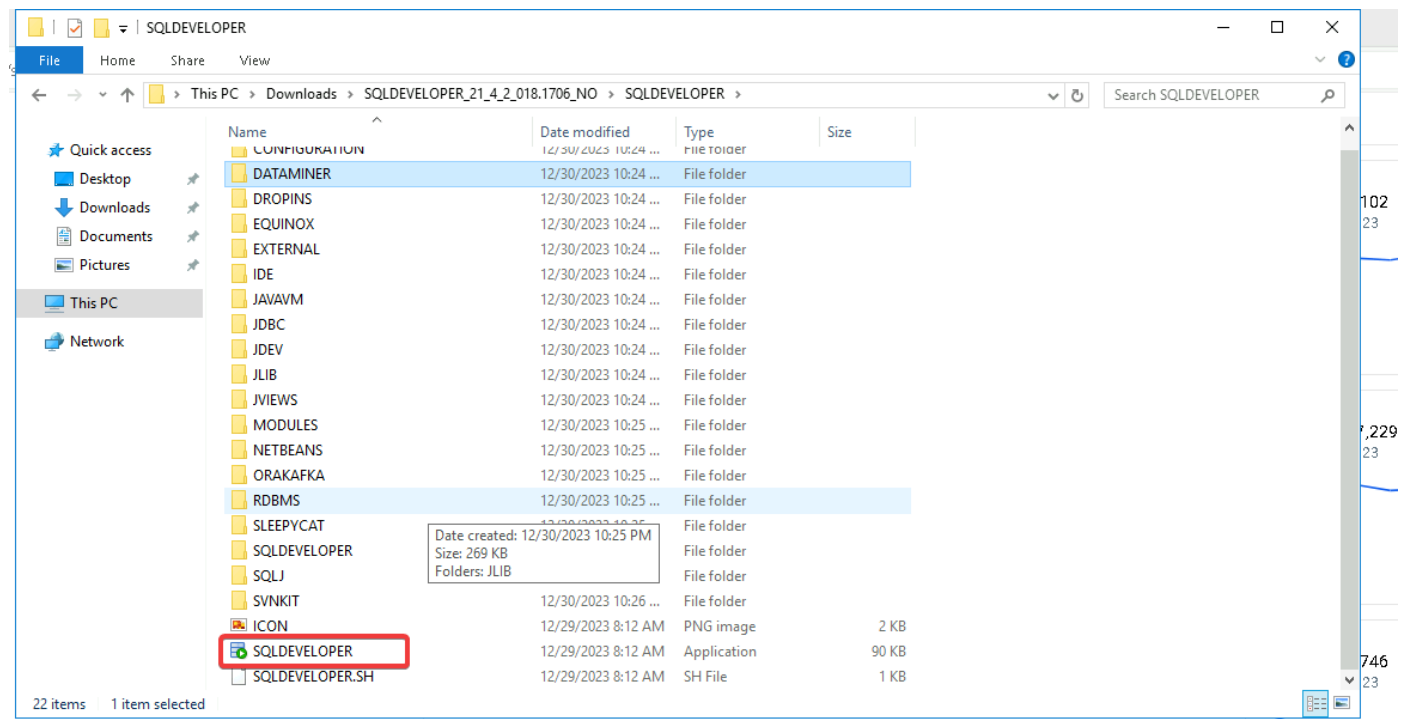
Total System Global Area 2449473096 bytes
Fixed Size                  9137736 bytes
Variable Size               536870912 bytes
Database Buffers            1895825408 bytes
Redo Buffers                 7639040 bytes
Database mounted.
Database opened.
SQL>

```

installing oracle sql developer

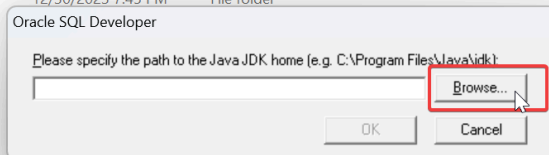
for sql developer to work you must install java jdk
first extract the sqldeveloper zip file .

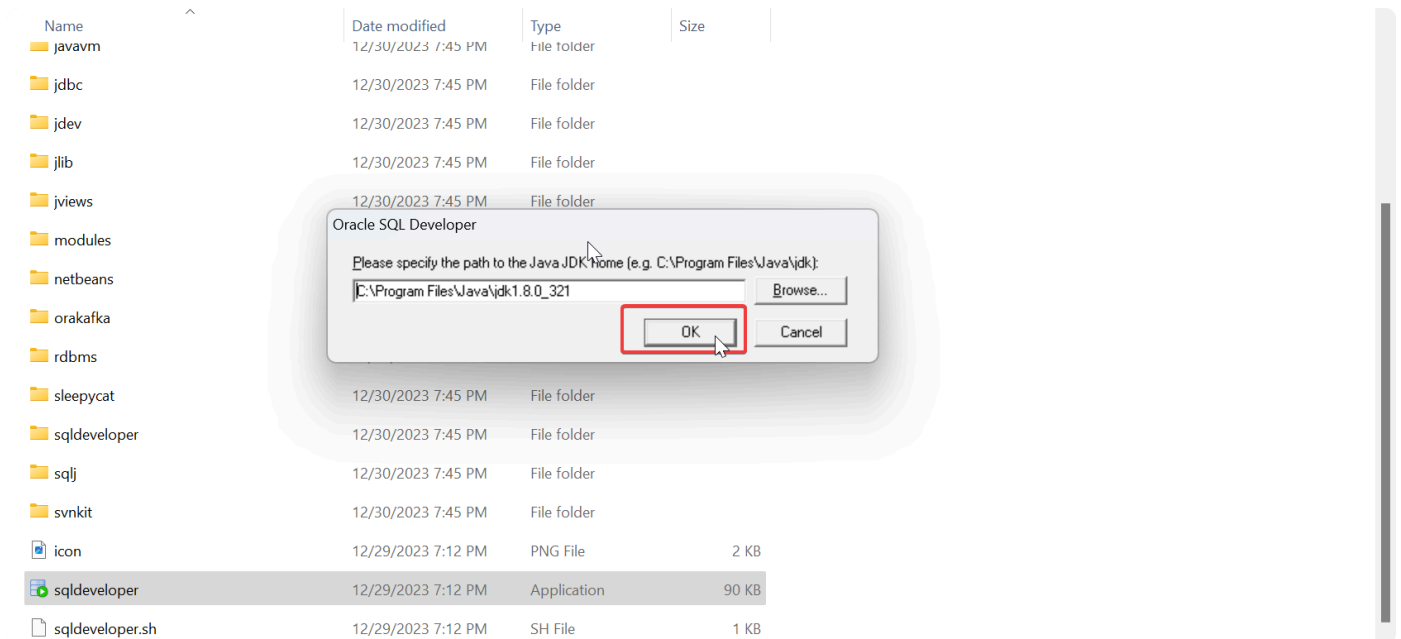
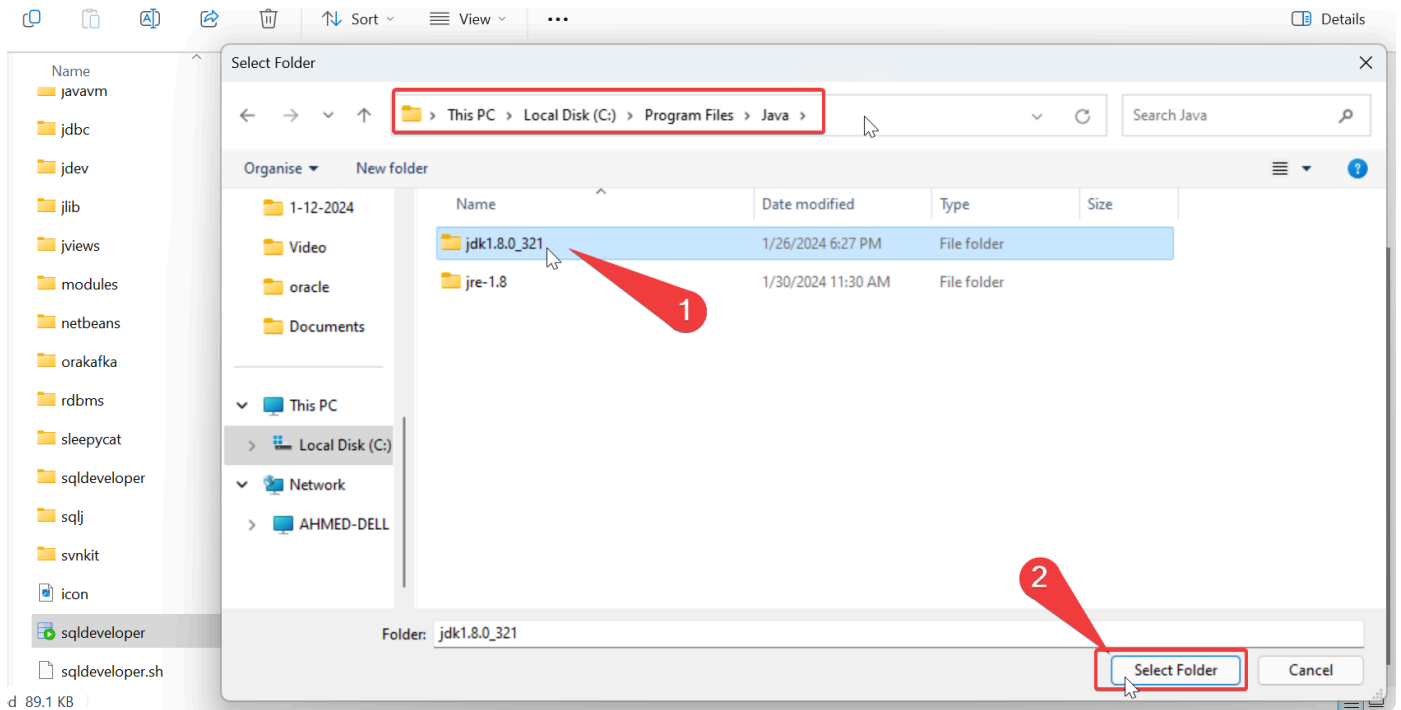
run the sqldeveloper.exe



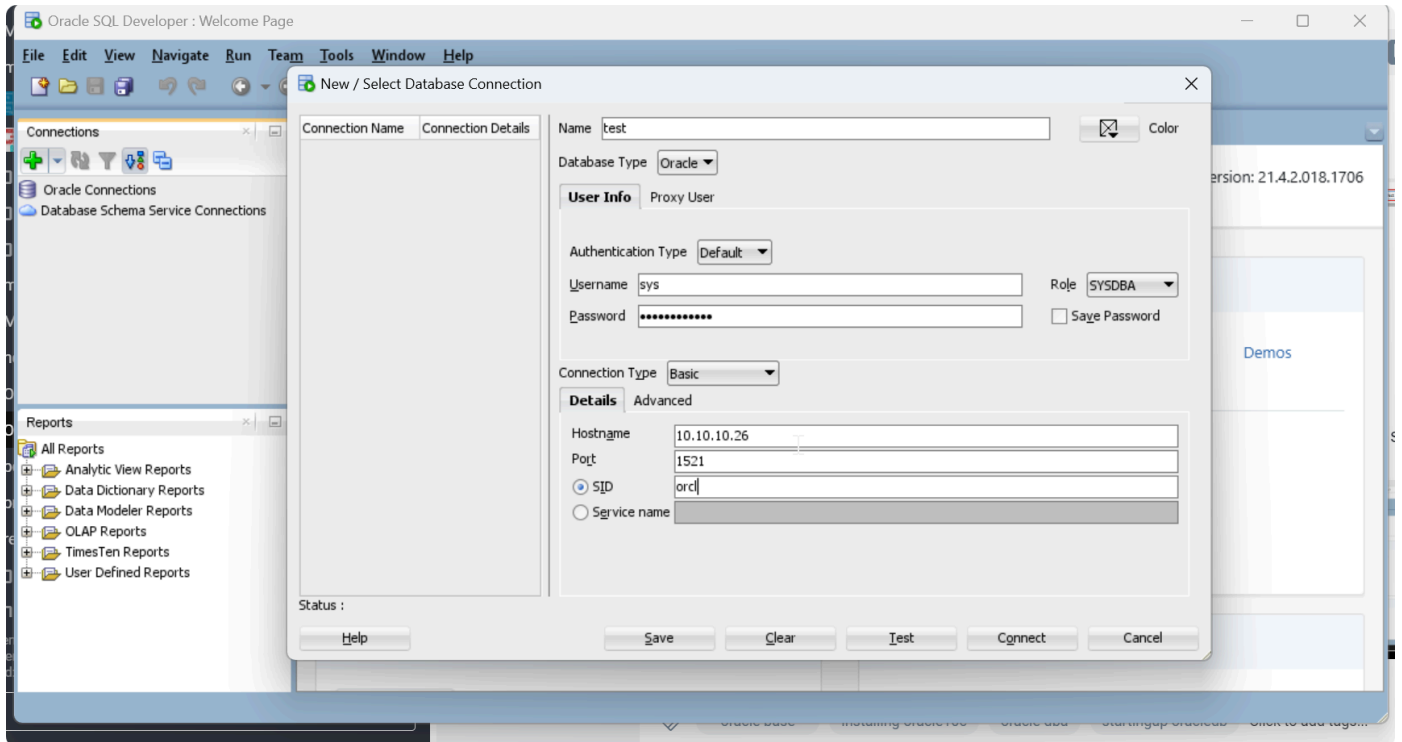
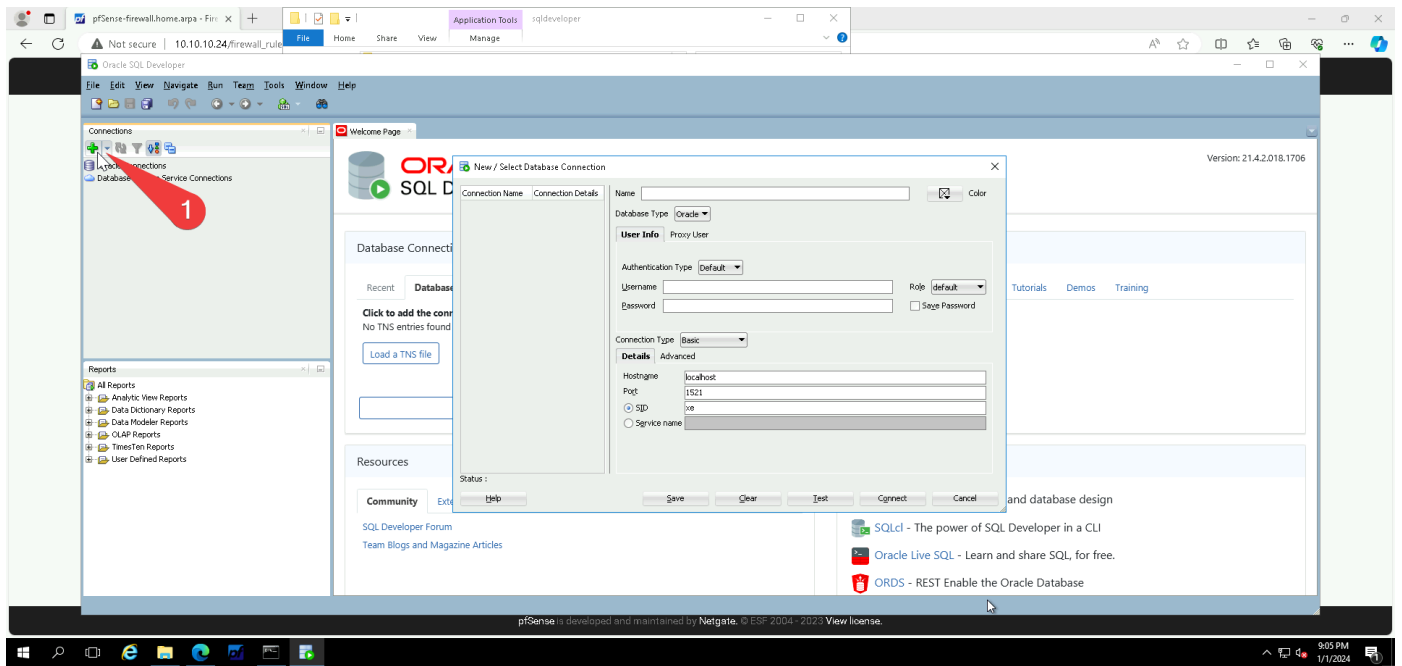
Name	Date modified	Type	Size
javavm	12/30/2023 7:45 PM	File folder	
jdbc	12/30/2023 7:45 PM	File folder	
jdev	12/30/2023 7:45 PM	File folder	
jlib	12/30/2023 7:45 PM	File folder	
jviews	12/30/2023 7:45 PM	File folder	
modules	12/30/2023 7:45 PM	File folder	
netbeans	12/30/2023 7:45 PM	File folder	
orakafka	12/30/2023 7:45 PM	File folder	
rdbms	12/30/2023 7:45 PM	File folder	
sleepycat	12/30/2023 7:45 PM	File folder	
sqldeveloper	12/30/2023 7:45 PM	File folder	
sqlj	12/30/2023 7:45 PM	File folder	
svnkit	12/30/2023 7:45 PM	File folder	
icon	12/29/2023 7:12 PM	PNG File	2 KB
sqldeveloper	12/29/2023 7:12 PM	Application	90 KB
sqldeveloper.sh	12/29/2023 7:12 PM	SH File	1 KB

Name	Date modified	Type	Size
javavm	12/30/2023 7:45 PM	File folder	
jdbc	12/30/2023 7:45 PM	File folder	
jdev	12/30/2023 7:45 PM	File folder	
jlib	12/30/2023 7:45 PM	File folder	
jviews	12/30/2023 7:45 PM	File folder	
modules	12/30/2023 7:45 PM	File folder	
netbeans	12/30/2023 7:45 PM	File folder	
orakafka	12/30/2023 7:45 PM	File folder	
rdbms	12/30/2023 7:45 PM	File folder	
sleepycat	12/30/2023 7:45 PM	File folder	
sqldeveloper	12/30/2023 7:45 PM	File folder	
sqlj	12/30/2023 7:45 PM	File folder	
svnkit	12/30/2023 7:45 PM	File folder	
icon	12/29/2023 7:12 PM	PNG File	2 KB
sqldeveloper	12/29/2023 7:12 PM	Application	90 KB
sqldeveloper.sh	12/29/2023 7:12 PM	SH File	1 KB





once the the application started click + sing to create connection



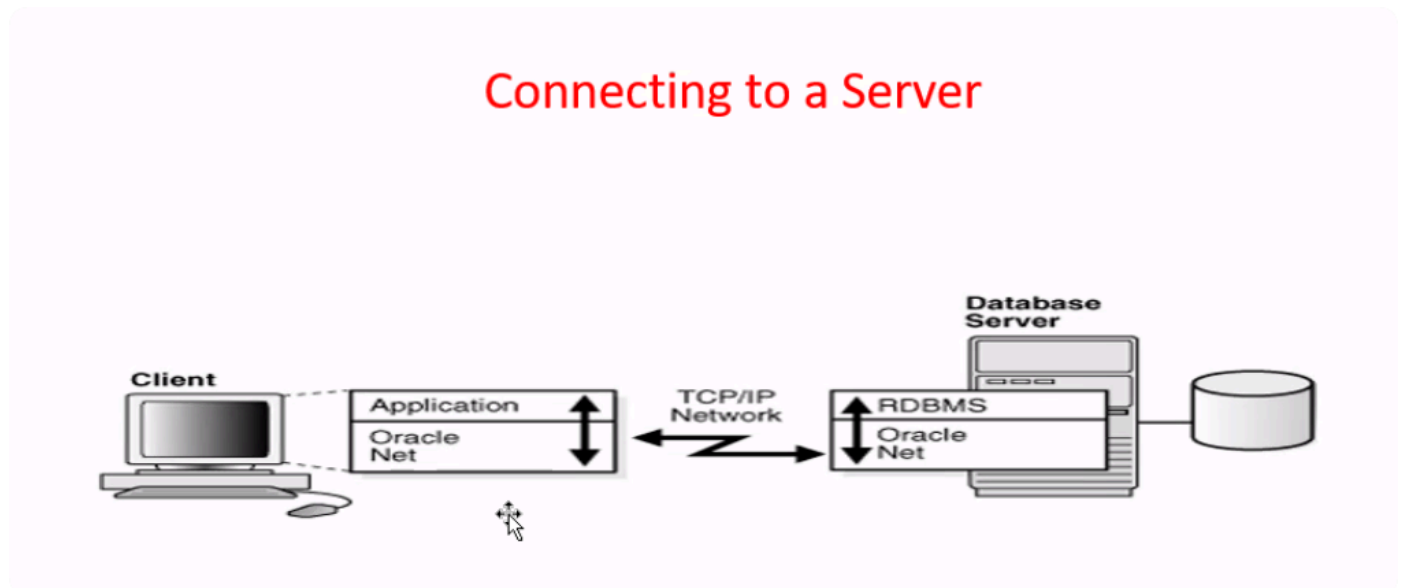
2. database engine and listener articture

- [connecting to database](#)
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Connecting to database

connect to database remotely :

if you have application server , in this case we will provide connection parameter to application server to connect to database . this similar to what we did with sql developer application



listener

is responsible for communication between the client and database engine , listener handle the authenticating by communicating with database engine and allow the privilege base on permission the user has

the config of the listener can be found in the below path

```

lsone dbjava diagnostics hr_install.log jdbc md OPatch oui R root.sh.old.1 sqlj use
rsr dbs dmu hs jdk mgw opmn owm racg runInstaller sqlpatch util

[oracle@oracledb share]$ cd /
[oracle@oracledb /]$ ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin share srv sys tmp u01 usr var
[oracle@oracledb /]$ cd u01/
[oracle@oracledb u01]$ ls
app oradata
[oracle@oracledb u01]$ cd oradata/
[oracle@oracledb oradata]$ ls
[oracle@oracledb oradata]$ ll -l
total 0
[oracle@oracledb oradata]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:19:45

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=oracledb.local)(PORT=1521)))
STATUS of the LISTENER
-----
Alias LISTENER
Version TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date 05-JAN-2024 10:31:38
Uptime 0 days 0 hr. 48 min. 7 sec
Trace Level off
Security ON: Local OS Authentication
SNMP OFF
Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File /u01/app/oracle/diag/tnslnr/oracledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=oracledb)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=oracledb)(PORT=5500))(Security=(my_wallet_directory=/u01/app/oracle/admin/orcl/xdw_wallet))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "orcl" status READY, has 1 handler(s) for this service...
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orcl", status READY, has 1 handler(s) for this service...
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orcl" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orclXDB" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orclpdb" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
The command completed successfully
[oracle@oracledb oradata]$ Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora

```

```

[oracle@oracledb oradata]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:19:45

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=oracledb.local)(PORT=1521)))
STATUS of the LISTENER
-----
Alias LISTENER
Version TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date 05-JAN-2024 10:31:38
Uptime 0 days 0 hr. 48 min. 7 sec
Trace Level off
Security ON: Local OS Authentication
SNMP OFF
Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File /u01/app/oracle/diag/tnslnr/oracledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=oracledb)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=oracledb)(PORT=5500))(Security=(my_wallet_directory=/u01/app/oracle/admin/orcl/xdw_wallet))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "orcl" status READY, has 1 handler(s) for this service...
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orcl", status READY, has 1 handler(s) for this service...
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orcl" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orclXDB" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orclpdb" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
The command completed successfully
[oracle@oracledb oradata]$ less /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
# listener.ora Network Configuration File: /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = oracledb.local)(PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
  )

/u01/app/oracle/product/19c/db_1/network/admin/listener.ora (END)

```

1. for host ip you want to connect
2. for the port its listening to
from listener you can change the default port , or change to which remote Ip listener should listen .

Changing the Listener Default Port Using Config File

you to go to config of the listener and change the port

```

[oracle@oracledb oradata]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:19:45

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=oracledb.local)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 19.0.0.0 - Production
Start Date                05-JAN-2024 10:31:38
UpTime                   0 days 0 hr. 48 min. 7 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnlsnr/oracledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=oracledb)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROCL1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=oracledb)(PORT=5500))(Security=(my_wallet_directory=/u01/app/oracle/admin/orcl/xdb_wallet))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "orcl" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
Service "orclpdb" has 1 instance(s).
  Instance "orcl", status READY, has 1 handler(s) for this service...
The command completed successfully

[oracle@oracledb oradata]$ less /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
# Listener.ora Network Configuration File: /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = oracledb.local)(PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
  )
/u01/app/oracle/product/19c/db_1/network/admin/listener.ora (END)

```

before you do that you have to stop the listener

```
lsnrctl stop
```

now vim to listener config and change the port

```

# Listener.ora Network Configuration File: /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = oracledb.local)(PORT = 5444))
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
  )

```

now start the listener

```
lsnrctl start
```

the listener is configured to accept connection on TCP 5444 port we defend

```

SNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:26:08

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting /u01/app/oracle/product/19c/db_1/bin/tnslsnr: please wait...

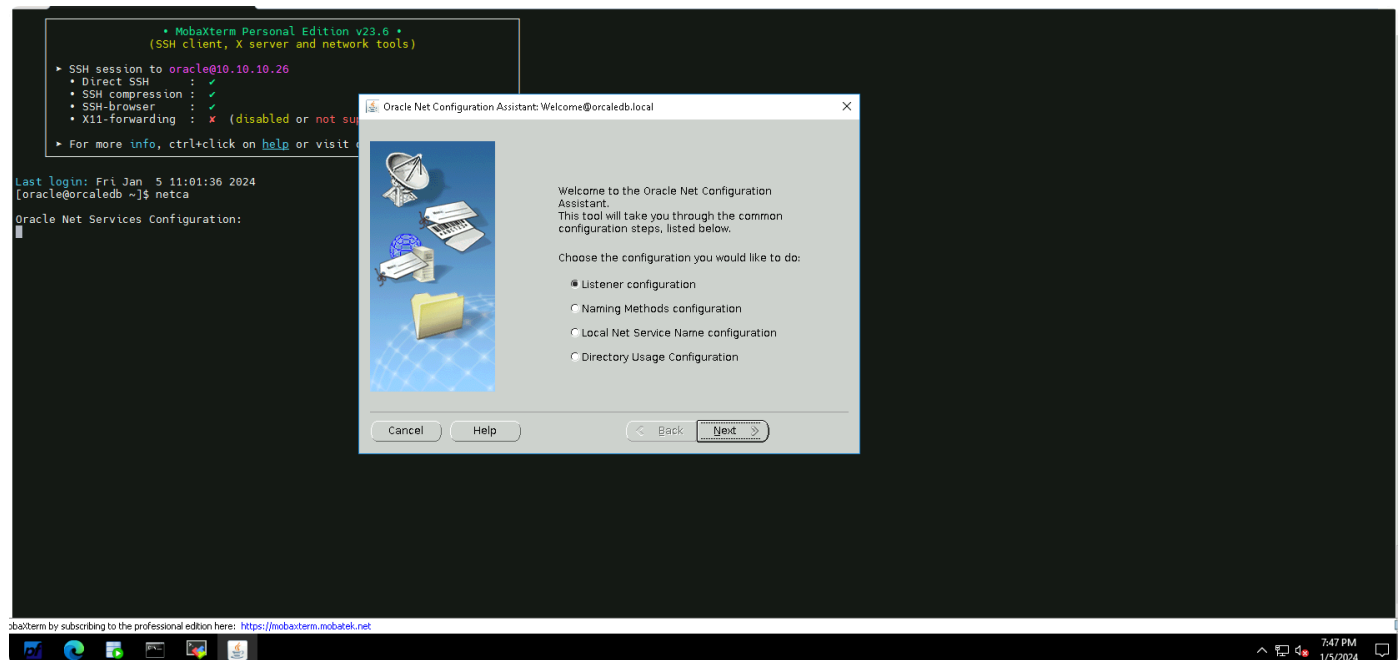
TNSLSNR for Linux: Version 19.0.0.0.0 - Production
System parameter file is /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Log messages written to /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=5444)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=5444)))
STATUS of the LISTENER
-----
Alias                LISTENER
Version              TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date           05-JAN-2024 11:26:08
uptime               0 days 0 hr. 0 min. 10 sec
Trace Level          off
Security              ON: Local OS Authentication
NMP                  OFF
Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File     /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=5444)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))
The listener supports no services
The command completed successfully
oracle@orcaledb oradata]$

```

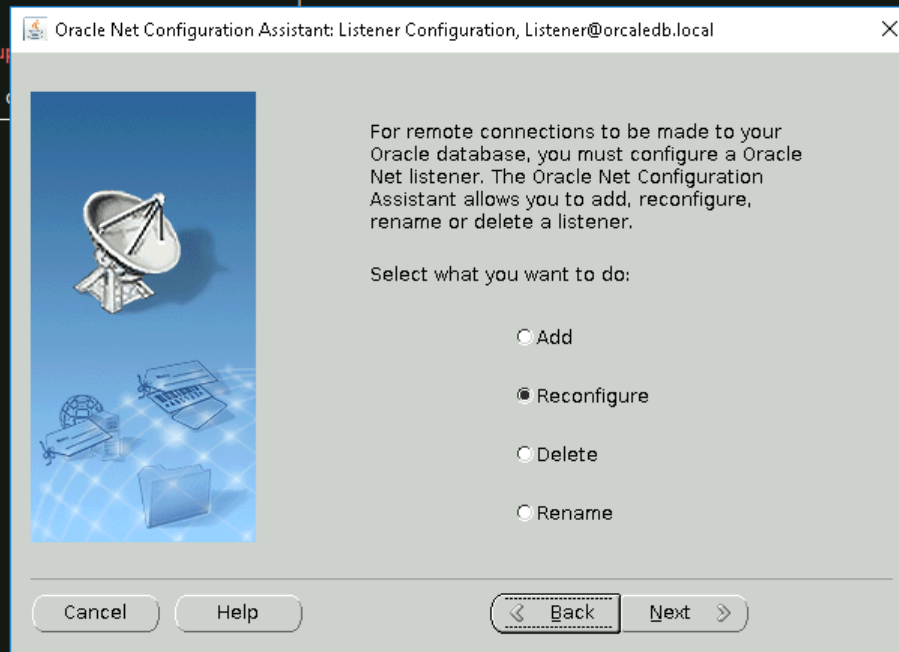
Changing the Listener Default Port Using Netca

netca is utility that help configure the listener of the to run this you have to run command from device that has java

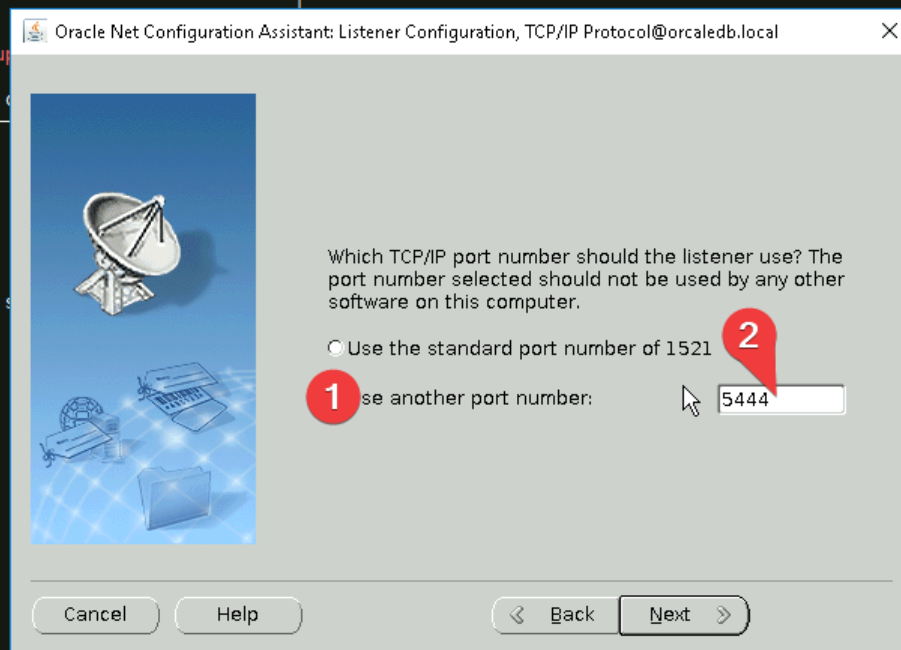


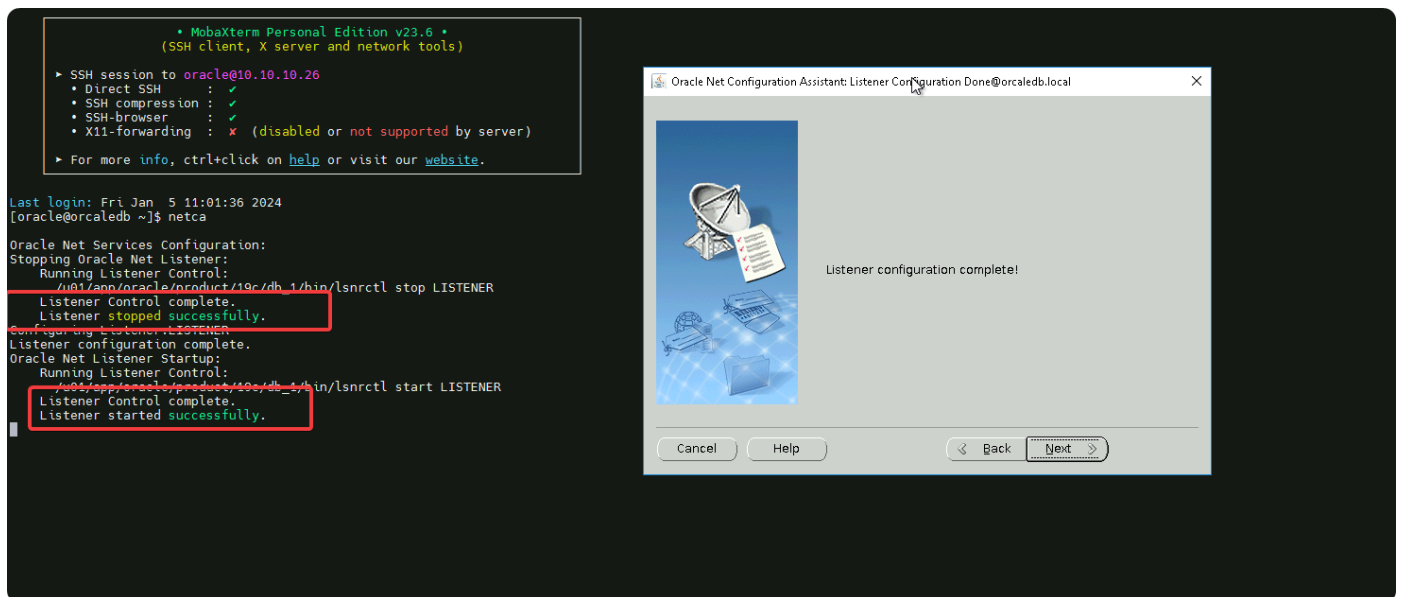
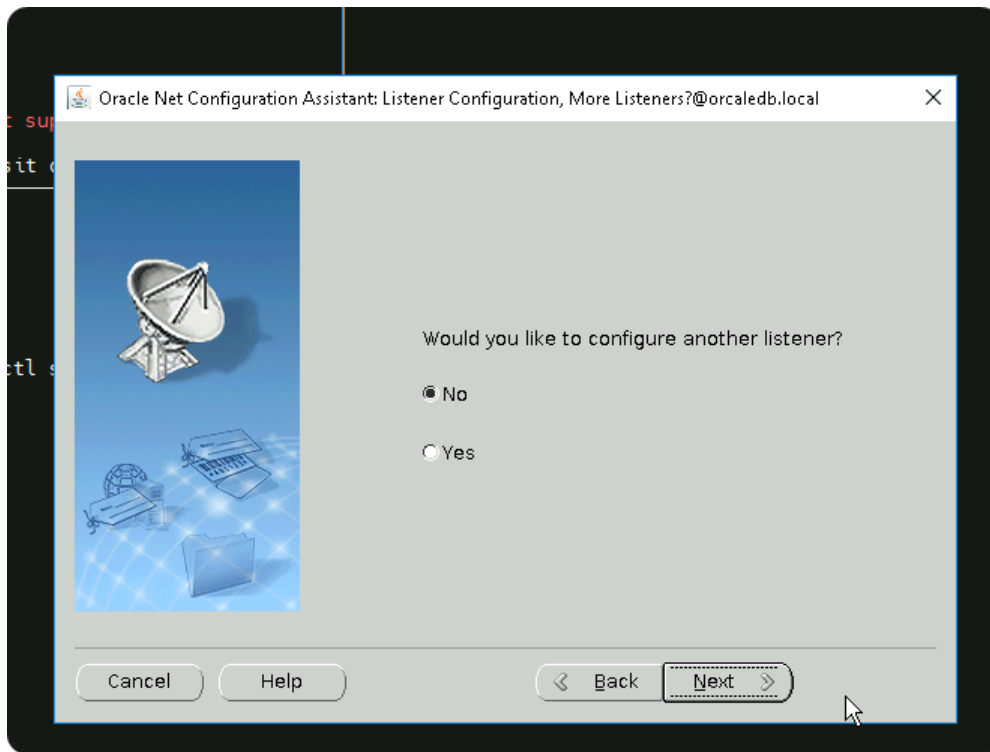
from the configuration you can make multitap listener or reconfigure the listener

for this purpose we will select reconfigure .



then it will ask the protocol keep it the same
next add the new port you want





the listener default port has been changed

```

Oracle Net Services Configuration:
Stopping Oracle Net Listener:
Running Listener Control:
/u01/app/oracle/product/19c/db_1/bin/lsnrctl stop LISTENER
Listener Control complete.
Listener stopped successfully.
Configuring Listener:LISTENER
Listener configuration complete.
Oracle Net Listener Startup:
Running Listener Control:
/u01/app/oracle/product/19c/db_1/bin/lsnrctl start LISTENER
Listener Control complete.
Listener started successfully.
Oracle Net Services configuration terminated by user. The exit code is -1
[oracle@orcaledb ~]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:51:43

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=5444)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                05-JAN-2024 11:50:42
Uptime                    0 days 0 hr. 1 min. 0 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslnr/orcaledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb.local)(PORT=5444)))
The listener supports no services
The command completed successfully
[oracle@orcaledb ~]$ lsnrctl start

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:51:52

Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01106: Listener using listener name LISTENER has already been started
[oracle@orcaledb ~]$

```

error after change the port

The screenshot shows the 'New / Select Database Connection' dialog in SQLcl. The 'Details' tab is active, showing the following configuration:

- Connection Name: con1
- Database Type: Oracle
- User Info: sys@//10.10.10.10...
- Authentication Type: Default
- Username: sys
- Password: [masked]
- Role: SYSDBA
- Connection Type: Basic
- Details:
 - Hostname: 10.10.10.26
 - Port: 5444
 - SID: orcl
 - Service name: [empty]

The status bar at the bottom indicates a failure: "Status : Failure -Test failed: Listener refused the connection with the following error:ORA-12505, TNS:listener does not currently know of SID given in connect descriptor (CONNECTION_ID=2B6HksHDSfq+1gueH...)"

```
/u01/app/oracle/product/19c/db_1/network/admin/
```

```
TNS-01106: Listener using listener name LISTENER has already been started
[oracle@orcaledb ~]$ cd /u01/app/oracle/product/19c/db_1/network/admin/
[oracle@orcaledb admin]$ ls
listener.ora  samples  shrept.lst  sqlnet.ora  tnsnames.ora
[oracle@orcaledb admin]$
```

the port has not changed we will change it and restart the listener and shut down and start the database

```
LISTENER_ORCL =
  (ADDRESS = (PROTOCOL = TCP)(HOST = orcaledb.local)(PORT = 1521))

ORCL =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = orcaledb.local)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl)
    )
  )
```

then stop and start the listner

```
lsnrctl stop
lsnrctl start
```

```
[oracle@orcaledb admin]$ lsnrctl stop

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:59:17

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=5444)))
The command completed successfully
[oracle@orcaledb admin]$ lsnrctl start

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 11:59:25

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting /u01/app/oracle/product/19c/db_1/bin/tnslsnr: please wait...

TNSLSNR for Linux: Version 19.0.0.0.0 - Production
System parameter file is /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Log messages written to /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=5444)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=5444)))
STATUS of the LISTENER
-----
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                05-JAN-2024 11:59:25
Uptime                    0 days 0 hr. 0 min. 10 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/orcaledb/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=5444)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=EXTPROC1521)))
The listener supports no services
The command completed successfully
[oracle@orcaledb admin]$
```

```
shutdown immediate
```

```
startup
```

```

The command completed successfully
[oracle@orcaledb admin]$ sqlplus sys/sys as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Jan 5 12:01:26 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup
ORACLE instance started.

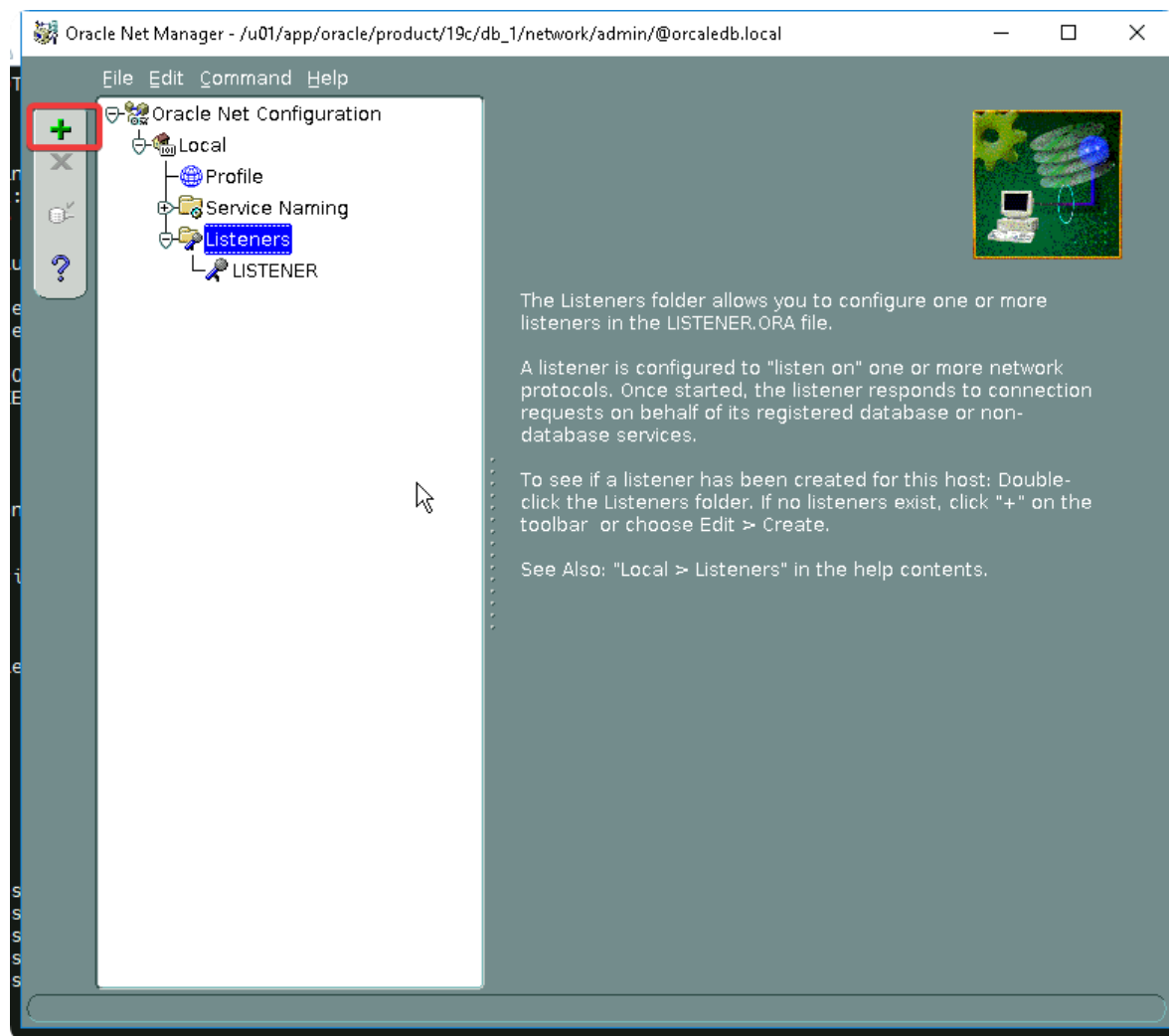
Total System Global Area 2449473096 bytes
Fixed Size          9137736 bytes
Variable Size       536870912 bytes
Database Buffers    1895825408 bytes
Redo Buffers        7639040 bytes
Database mounted.

```

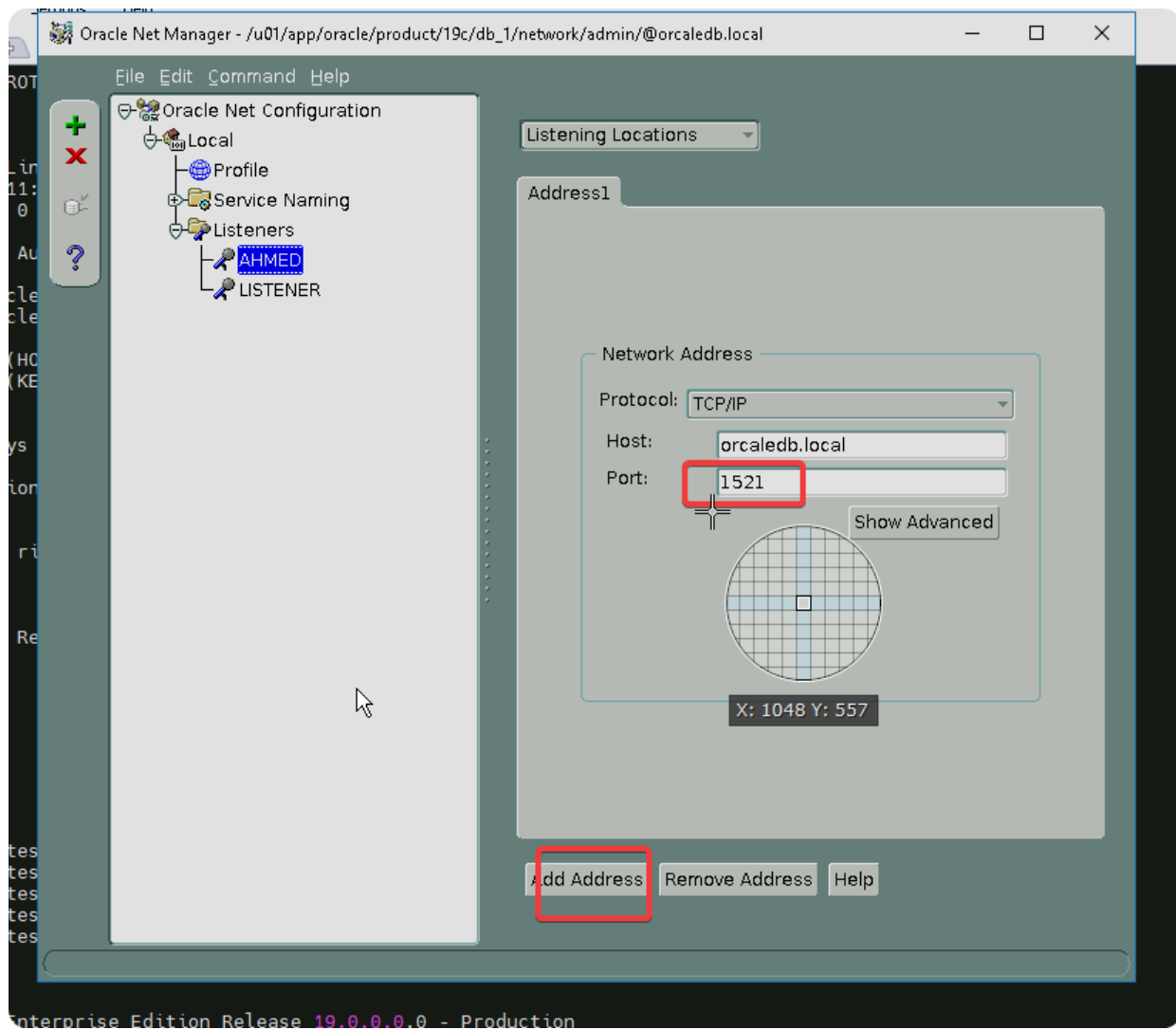
Oracle has a link to the Oracle Cloud Edition Page: <https://www.oracle.com/cloud/>

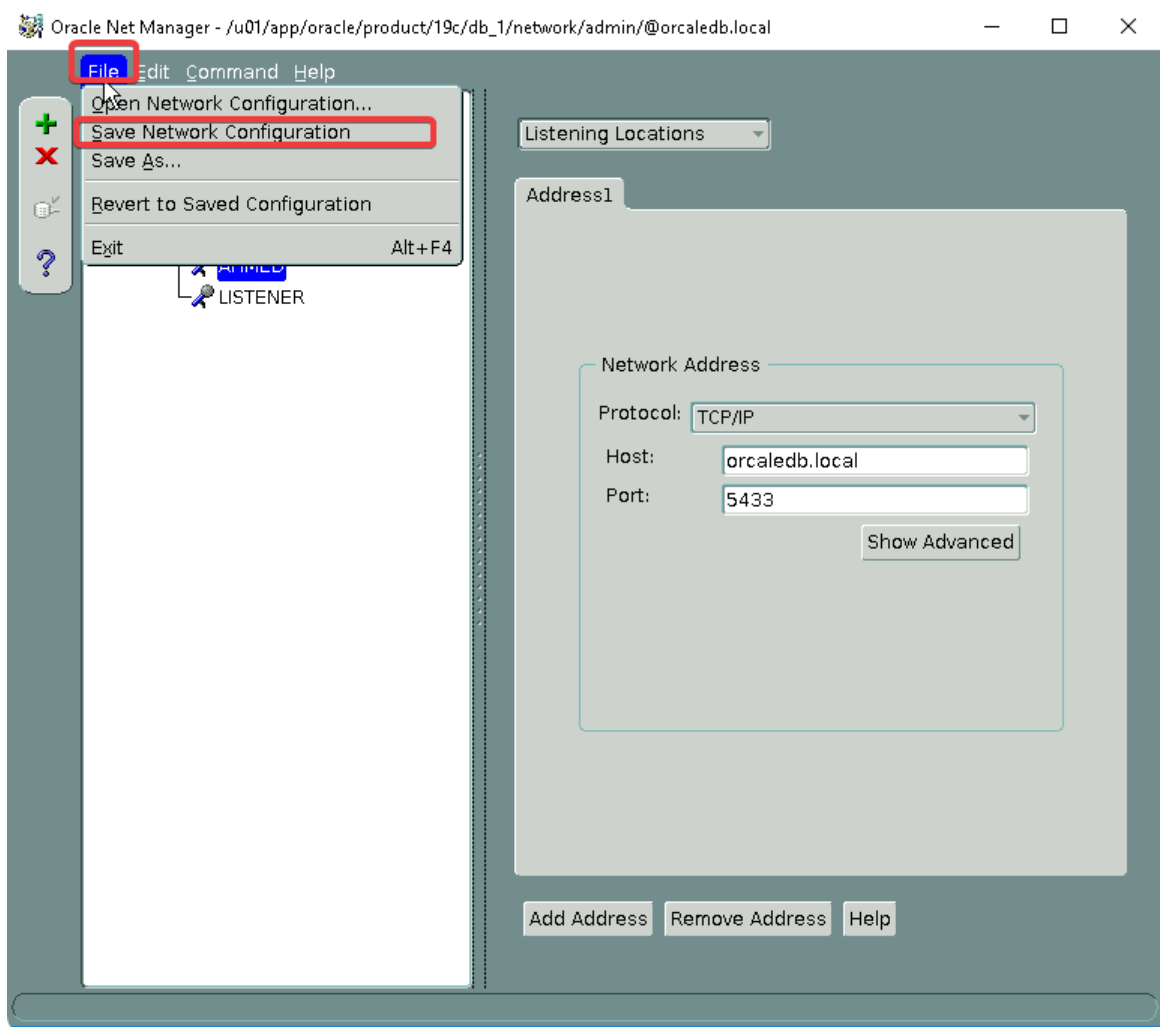
add more listener port netmgr

if you have multiple interface you can configure multiple listener to acuminate the interface , this is done by using netmgr .
the window will open expand the window till you reach listener and click + green sing



give friendly name
then click add address and then give random port and then click ok





```
lsnrctl start ahmed
```

```
[oracle@orcaledb admin]$ lsnrctl status ahmed
LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 05-JAN-2024 12:23:55
Copyright (c) 1991, 2019, Oracle. All rights reserved.
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=orcaledb.local)(PORT=5433)))
STATUS of the LISTENER
-----
Alias                ahmed
Version              TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date            05-JAN-2024 12:22:37
Uptime                0 days 0 hr. 1 min. 17 sec
Trace Level           off
Security              ON: Local OS Authentication
SNMP                  OFF
Listener Parameter File /u01/app/oracle/product/19c/db_1/network/admin/listener.ora
Listener Log File      /u01/app/oracle/diag/tnslsnr/orcaledb/ahmed/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=orcaledb)(PORT=5433)))
The listener supports no services
The command completed successfully
[oracle@orcaledb admin]$
```

oracle database architecture

RAM architecture

oracle ram is divided to two

1. sga system global area

is the space that occupied in memory in order to start the database engine
inside it we will find this component to run the database engine

- database buffer cache
- redo log buffer cache
- archive log

2. pga Program Global Area

It is a memory area that contains a couple of sub-areas in it. Unlike the SGAs, PGAs are private for each user. So any other user cannot see or touch that memory area.

In PGA, there are a couple of sub-memory areas. As you can see in the diagram

pga is responsible to open connection between the client and database .
it use the listener to open the connection

shared pool

it response of taking the query and execute the query on data folder and return the result

large pool

similar to shared pool but specified for large query's with loops inside of it .

buffer cache

this is responsible for keep store of result of common executed query to avoid going to the data folder to speed up the process of query executing , also reduce CPU usage .

the query that goes to shared pool to be executed , if the query is getting executed many time , then shared pool will send the query to buffer cache to keep the result of the query to save time and load .

java pool

responsible for java command & query's

redo log buffer

if you have done update on row but you have not commit , redo log will take copy of the original row in case another update is running on the same row

redo log allow you to rollback

redo log will be done once we commit the transaction.

stream pool

reasonable for query contain any media such as photos or videos.

3. managing database instance paramter

- [how to view the parameters](#)
- [preciouses to be follow before editing the parameter](#)
 - [taking bckup of psfile](#)
- [how to edit the parameter](#)
 - [basic parameter](#)
 - [advanced parameter](#)
- [example 1change java pool parameter.](#)
- [rollback the configuration](#)

the parameter for configuring the instance such as for spa & pga are found on the below files .

"spfileorcl.ora" & "initorcle.ora"

this both have all parameter that defined how database run .

there are many parameter in the file you can edit

its not recommended to edit the parameters by opening the "spfileorcl.ora" & "initorcle.ora" as the database may not work after restart

how to view the parameters

to view the parameters configured on database login to oracle and execute the below command

```
show parameters;
```

```
SQL> show parameters;
```

NAME	TYPE	VALUE

DBFIPS_140	boolean	FALSE
active_instance_count	integer	
adg_account_info_tracking	string	LOCAL
adg_redirect_dml	boolean	FALSE
allow_global_dblinks	boolean	FALSE
allow_group_access_to_sga	boolean	FALSE
allow_rowid_column_type	boolean	FALSE
approx_for_aggregation	boolean	FALSE
approx_for_count_distinct	boolean	FALSE
approx_for_percentile	string	none
aq_tm_processes	integer	1

NAME	TYPE	VALUE

archive_lag_target	integer	0
asm_diskstring	string	
asm_preferred_read_failure_groups	string	
audit_file_dest	string	/u01/app/oracle/admin/orcl/adu mp
audit_sys_operations	boolean	TRUE
audit_syslog_level	string	
audit_trail	string	DB
autotask_max_active_pdbs	integer	2
awr_pdb autoflush_enabled	boolean	FALSE
awr_pdb_max_parallel_slaves	integer	10

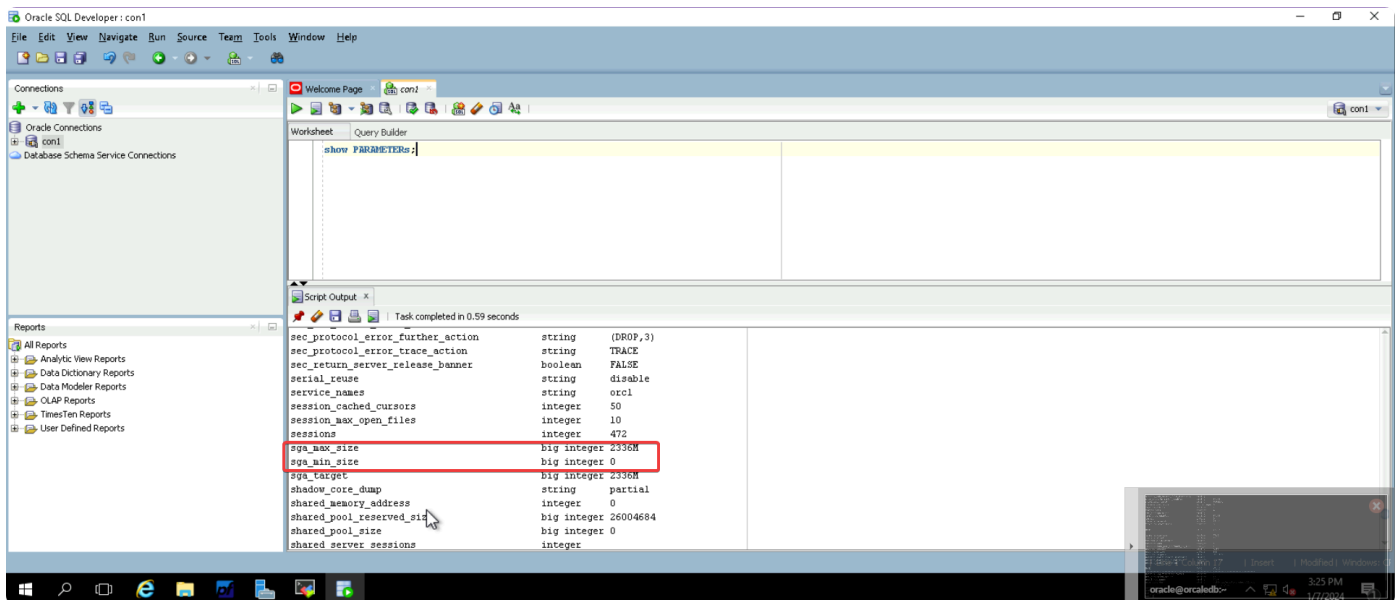
NAME	TYPE	VALUE

also you can view the parameter from sql developer for better view of the parameter just connect to oracle and issue the same command .

you will find the path for "spfileorcl.ora" & "initoracle.ora" & also you will find parameters for large pool and spa pga

The screenshot displays the Oracle SQL Developer interface. The main window shows the output of the 'show parameters;' command. The 'large_pool_size' parameter is highlighted with a red box, indicating its value is 0. The 'job_queue_processes' parameter is also visible, with a value of 40. The interface includes a 'Connections' pane on the left, a 'Script Output' pane at the bottom, and a 'Reports' pane on the right.

Parameter Name	Type	Value
job_queue_processes	integer	40
large_pool_size	big integer	0
trap_directory_access	string	none
ldap_directory_sysauth	string	no
license_max_sessions	integer	0
license_max_users	integer	0
license_sessions_warning	integer	0
listener_networks	string	
lob_signature_enable	boolean	FALSE
local_listener	string	LISTENER_ORCL
lock_name_space	string	
lock_sga	boolean	FALSE
log_archive_config	string	
log_archive_dest	string	
log_archive_dest_1	string	
log_archive_dest_10	string	



```

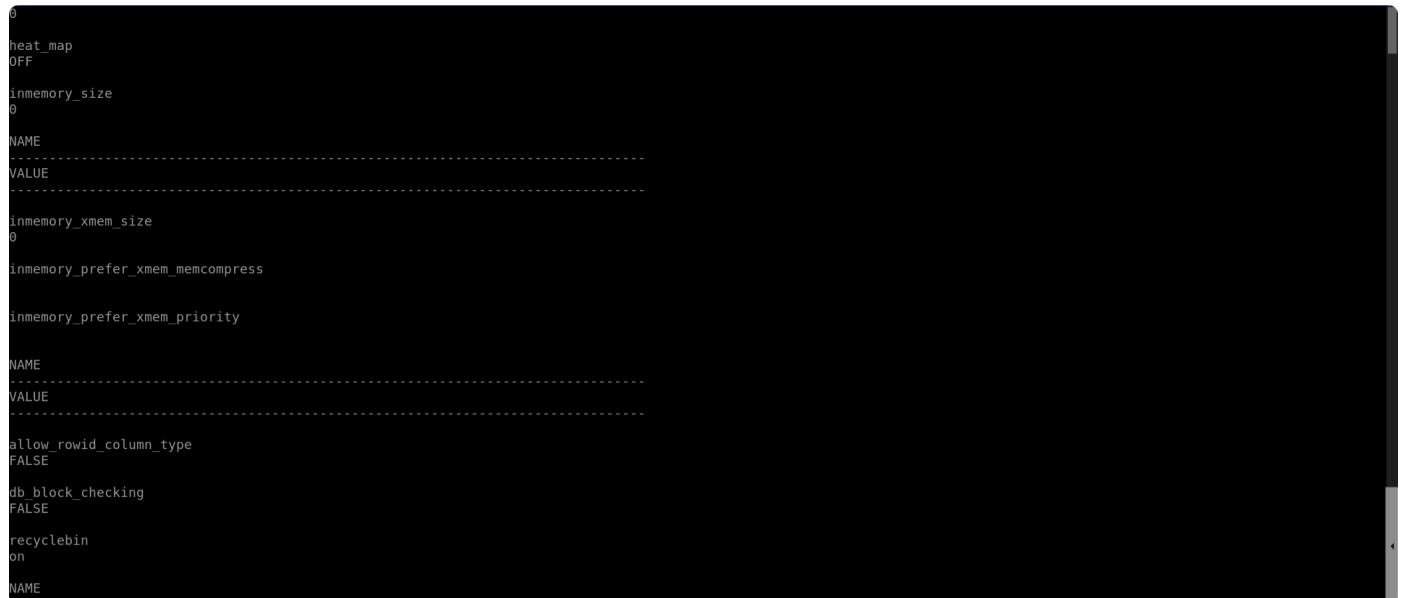
spfile                                string    /u01/app/oracle/product/19c/db_1/dbs/spfileorcl.ora
sga_max_size                          big integer 2336M

```

some parameters has dependency such sga pga , you cannot change their parameters with informing the parameter" memory target "

for better view of the parameter
use the below command .

```
select name , value from v$parameter;
```



The screenshot shows the SQL Developer interface. The top pane displays the query: `select name , value from v$parameter;`. The bottom pane shows the query results, which are 50 rows fetched in 0.266 seconds. The results are displayed in a table with two columns: NAME and VALUE.

	NAME	VALUE
1	lock_name_space	{null}
2	processes	300
3	sessions	472
4	timed_statistics	TRUE
5	timed_os_statistics	0
6	resource_limit	TRUE
7	license_max_sessions	0
8	license_sessions_warning	0
9	long_module_action	TRUE
10	standby_db_preserve_states	NONE
11	instance_abort_delay_time	0
12	cpu_count	2
13	cpu_min_count	2

```
select name , value from v$parameter where name =('spfile');
```

```
SQL> select name , value from v$parameter where name =('spfile');
NAME
-----
VALUE
-----
spfile
/u01/app/oracle/product/19c/db_1/dbs/spfileorcl.ora

SQL>
```

preciouses to be follow before editing the parameter

before editing the parameter need to have rollback plan in case of misconfiguration

you need to take backup of "spfileorcl.ora" & "initorcl.ora" , you can do it from oracle sql cmd

taking backup of psfile

```
create pfile='path for where you want save the spfile/filename.ora' from spfile;
create pfile ='share/pfile.ora' from spfile;
```

```
SQL> create pfile = '/share/pfile.ora' from spfile;
File created.
SQL> █
```

```
oracle@orcaledb ~]$ cd /share/
oracle@orcaledb share]$ ls
addnode      ctx           deinstall    has           jdbc          mgw           oracore      pfile.ora    relnotes      slax          ucp
apex         cv            demo         hr_install.log jdk           network      ord          ptsql        root.sh       sqldeveloper  usm
assistants  data         diagnostics  hs            jlib          nls          ords         precomp     root.sh.old  sqlj          utl
bin          dbjava       dmu          install       ldap          odbc         oss          QOPatch     root.sh.old.1 sqlpatch      wwg
clone       dbs          drdaas       instantclient lib           olap         oui          R            runInstaller sqlplus      xdk
crs         db-sample-schemas-23.2
css         db-sample-schemas-23.2.zip
oracle@orcaledb share]$ █
```

how to edit the parameter

you will use the command `alter` follow by parameter name and parameter value .

the parameter it self are divided to two

basic parameter

this are easy to edit parameter such as 'control_file' , 'db_block_size' , 'undo_tablespace' and many more .

you can change the parameter and continue working on database.
changes here doesn't require restart the instance .

advanced parameter

this parameter has some specific steps such as 'shared_pool_size' 'db_cach_size' this parameters are specific for core component of the database engine .

the parameters here will not take effect unless you restart the instance .

example 1- change java pool parameter.

https://docs.oracle.com/en/database/oracle/oracle-database/19/refrn/JAVA_POOL_SIZE.html#GUID-F24C4770-0B7C-42BD-9AAD-0EC1E92A90F6

check the link to know about the parameter and the allowed values

Java session space at end-of-call.

Property	Description
Parameter type	Big integer
Syntax	<code>JAVA_POOL_SIZE = integer [K M G]</code>
Default value	If <code>SGA_TARGET</code> is set: If the parameter is not specified, then the default is 0 (internally determined by the Oracle Database). If the parameter is specified, then the user-specified value indicates a minimum value for the memory pool. If <code>SGA_TARGET</code> is not set: 24 MB, rounded up to the nearest granule size
Modifiable	ALTER SYSTEM
Modifiable in	No

the syntax is as follow

```
alter system set [parameter name] = [value ] scope=[both] or [spfile]
```

```
alter system set java_pool_size='151' scope=both;
```

```
SQL> alter system set java_pool_size='151' scope=both;
System altered.
SQL> █
```

```
shutdown immediate ;

startup ;
```

```
SQL> alter system set java_pool_size='151' scope=both;

System altered.

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup
ORACLE instance started.

Total System Global Area 2449473096 bytes
Fixed Size                  9137736 bytes
Variable Size              553648128 bytes
Database Buffers          1879048192 bytes
Redo Buffers                7639040 bytes
Database mounted.
Database opened.
SQL> █
```

```
select name , value from v$parameter where name =('java_pool_size');
```

```
SQL> select name , value from v$parameter where name =('java_pool_size');

NAME
-----
VALUE
-----
java_pool_size
16777216

SQL> █
```

note : the value is displayed in bites.

rollback the configuration

in case you changed the parameter and the database is not starting up , the you can rollback using the backup spfile you created before you edited the parameter.

first shutdown the database

```
shutdown immediate
```

then startup the database using the spfile you took copy of it

```
startup pfile='/share/pfile.ora';
```

```
SQL> startup pfile='/share/pfile.ora';  
ORACLE instance started.
```

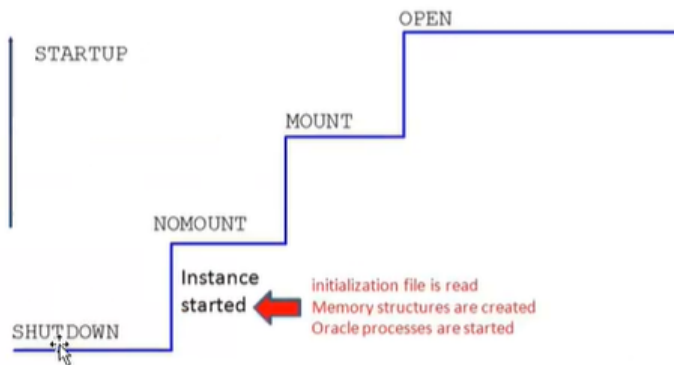
```
Total System Global Area 2449473096 bytes  
Fixed Size                  9137736 bytes  
Variable Size               536870912 bytes  
Database Buffers            1895825408 bytes  
Redo Buffers                 7639040 bytes  
Database mounted.  
Database opened.  
SQL> █
```


4. database mode

- [startup modes](#)
- [shutdown mode](#)

startup modes

the database mode while starting



1- **shutdown** :when you startup the server the database will be in shutdown state

2.**nomount** : when you execute the `startup` command the instance start will begin and then database will move to nomount, in the **nomount** this will happen

- read the parameter file (spfile) , and **init file**.
- memory is allocated in pfile
- background process in spfile and init file

3- **mount** : next the database will move to mount file which include the below steps

- read the control file
- datafile will be visible but not available to users
- the recovery stage , which include the status of database before shutdown .

4-**open** : in this mode the database will be available to users to use the database .following will be done during this mode

-data file integrity is verified

```

Last login: Mon Jan 8 01:55:00 2024 from 10.10.10.4
[oracle@orcaledb ~]$ sqlplus sys/sys as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 8 04:39:49 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> shutdown
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup
ORACLE instance started.

Total System Global Area 2449473096 bytes
Fixed Size 9137736 bytes
Variable Size 553648128 bytes
Database Buffers 1879048192 bytes
Redo Buffers 7639040 bytes
Database mounted.
Database opened.
SQL>

```

nomount mode

mount mode

X: 1113 Y: 483

open

shutdown mode

there are many type of shutdown

Shutdown Mode	A	I	T	N
Allows new connections	No	No	No	No
Waits until current sessions end	No	No	No	Yes
Waits until current transactions end	No	No	Yes	Yes
Forces a checkpoint and closes files	No	Yes	Yes	Yes

Shutdown modes:

- **A = ABORT**
- **I = IMMEDIATE**
- **T = TRANSACTIONAL**
- **N = NORMAL**

Immediate Shutdown

Normal Shutdown: Think of this as the polite way to shut down Oracle. In normal shutdown mode, the database doesn't accept new connections, but it waits for existing transactions to complete before closing. It's like letting everyone finish their conversations before turning off the lights.

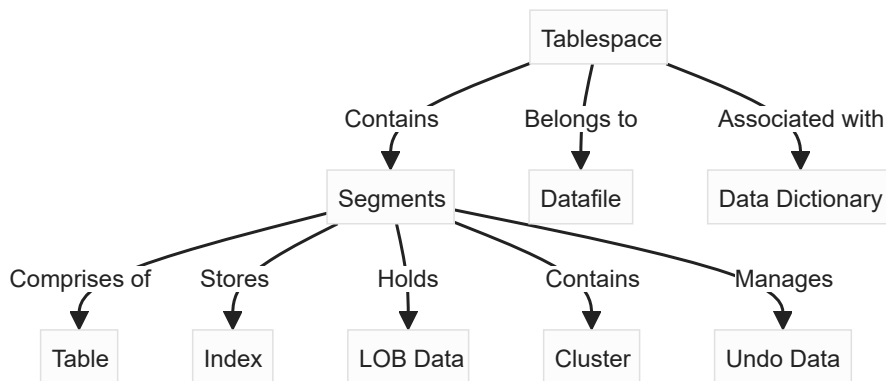
-Transactional Shutdown: This mode is even more patient. It allows all transactions to complete, and then it shuts down the database gracefully. It's like waiting for the last guest to leave the party before cleaning up.

- **Immediate Shutdown Abort:** This one is a bit like yanking the power cord out of your computer. It forcefully terminates the Oracle database without any regard for ongoing transactions. It's not the recommended way to do it, but sometimes it's necessary in extreme situations.

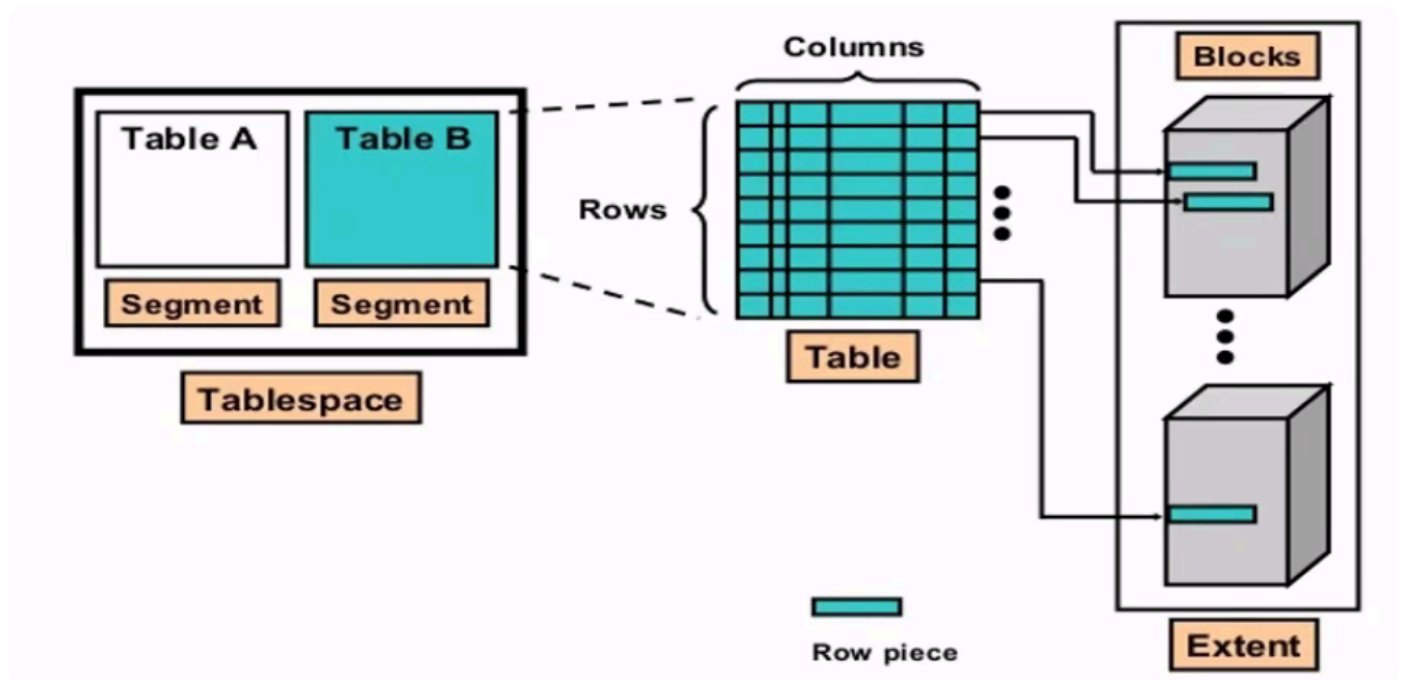
5.managed database storage structure

database engine will not talk directory with storage of server when it need to store data in the disk

tablespace

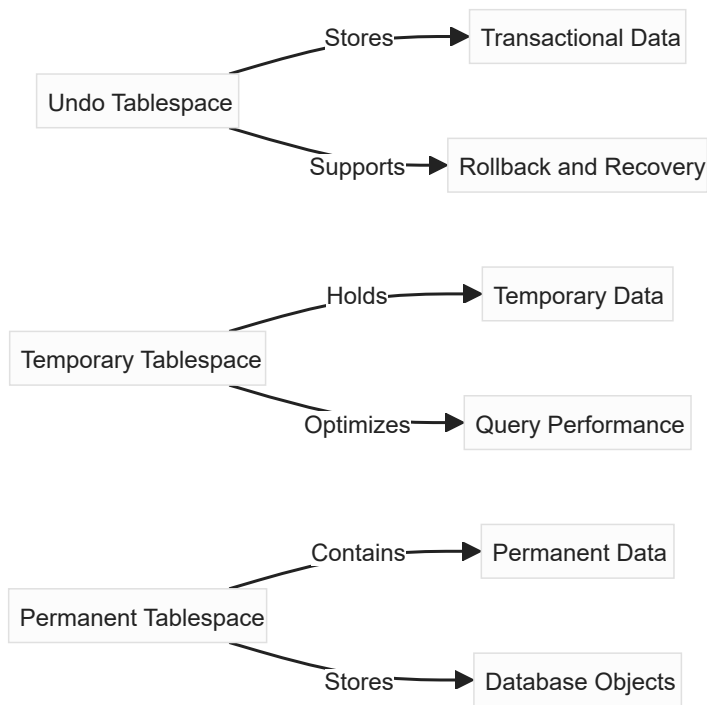


is place where the tables and data will be stored



segment

tablespace type



undo tablespace

The redo log employs this tablespace to store transaction data that has not yet been committed. This ensures that in the event of a rollback, the necessary information is readily available, offering a safeguard against data inconsistencies or incomplete transactions.

temporary tablespace

The temporary tablespace is harnessed by the buffer pool to house the results of queries that are frequently executed. It acts as a short-term storage repository for this data, which is eventually purged after its temporary use, ensuring efficient memory management.

permanent tablespace

A 'Permanent Tablespace' is where all the important stuff in a database lives – like tables and their data. It's the safe and stable home for your valuable information, and it sticks around even when you turn off the computer or close the application. Just remember, when you want your data to stay put, it goes into the Permanent Tablespace.

creating segmented of tablespace for users

In the context of an Oracle database with multiple concurrent users, there exists a potential challenge where the shared tablespace might reach its storage capacity due to the cumulative data generated by these users. To pre-emptively address this issue and optimize data management, a common strategy involves the creation of distinct tablespaces for each individual user. You can also consider increasing the space for tablespace but for better management creating separate tablespace for each user is considered a better and best practice.

first use the below command to check all tablespace in database

```
select tablespace_name , file_name from dba_data_files;
```

```

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 12-JAN-2024 10:38:17
Copyright (c) 1991, 2019, Oracle. All rights reserved.

TNS-01106: Listener using listener name LISTENER has already been started
[oracle@oracledb ~]$ sqlplus sys/sys as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Jan 12 10:38:55 2024
Version 19.3.0.0.0
Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> create user ahmed identified by 101124738;
create user ahmed identified by 101124738
+
ERROR at line 1:
ORA-65096: invalid common user or role name

SQL> select tablespace_name , file_name from dba_data_files;

TABLESPACE_NAME
-----
FILE_NAME
-----
USERS
/u01/app/oracle/oradata/ORCL/users01.dbf
UNDOTBS1
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
SYSTEM
/u01/app/oracle/oradata/ORCL/system01.dbf

TABLESPACE_NAME
-----
FILE_NAME
-----
SYSAUX
/u01/app/oracle/oradata/ORCL/sysaux01.dbf

SQL> █

```



creating tablespace

use the below syntax when creating tablespace

syntax:

```

CREATE TABLESPACE tablespace_name
DATAFILE 'datafile_name.dbf' SIZE size;

```

```
create tablespace tbs1 datafile 'tbs1_data.dbf' size 100m;
```

```

SQL> create tablespace tbs1 datafile 'tbs1_data.dbf' size 100m;
tablespace created.

SQL> █

```

the tablespace is stored in directory **oradata**

we can use the below query to get the path for tablespace we just created .

```
select tablespace_name , file_name from dba_data_files;
```

```

USERS
/u01/app/oracle/oradata/ORCL/users01.dbf

UNDOTBS1
/u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYSTEM
/u01/app/oracle/oradata/ORCL/system01.dbf

TABLESPACE_NAME
-----
FILE_NAME
-----
SYSaux
/u01/app/oracle/oradata/ORCL/sysaux01.dbf

SQL> create tablespace tbs1 datafile 'tbs1_data.dbf' size 100m;
Tablespace created.

SQL> select tablespace_name , file_name from dba_data_files;

TABLESPACE_NAME
-----
FILE_NAME
-----
USERS
/u01/app/oracle/oradata/ORCL/users01.dbf

UNDOTBS1
/u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYSTEM
/u01/app/oracle/oradata/ORCL/system01.dbf

TABLESPACE_NAME
-----
FILE_NAME
-----
SYSaux
/u01/app/oracle/oradata/ORCL/sysaux01.dbf

TBS1
/u01/app/oracle/product/19c/db_1/dbs/tbs1_data.dbf

SQL>

```

```

SYSTEM
/u01/app/oracle/oradata/ORCL/system01.dbf

TABLESPACE_NAME
-----
FILE_NAME
-----
SYSaux
/u01/app/oracle/oradata/ORCL/sysaux01.dbf

SQL> create tablespace tbs1 datafile 'tbs1_data.dbf' size 100m;
Tablespace created.

SQL> select tablespace_name , file_name from dba_data_files;

TABLESPACE_NAME
-----
FILE_NAME
-----
USERS
/u01/app/oracle/oradata/ORCL/users01.dbf

UNDOTBS1
/u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYSTEM
/u01/app/oracle/oradata/ORCL/system01.dbf

TABLESPACE_NAME
-----
FILE_NAME
-----
SYSaux
/u01/app/oracle/oradata/ORCL/sysaux01.dbf

TBS1
/u01/app/oracle/product/19c/db_1/dbs/tbs1_data.dbf

SQL> quit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@oracledb ~]$ cd /u01/app/oracle/product/19c/db_1/dbs/
[oracle@oracledb dbs]$ ls
hc_orcl.dat hc_worlber.dat init.ora lkorcl lkWORLDBER orapworcl orapworlber spfileorcl.ora spfileworlber.ora tbs1_data.dbf
[oracle@oracledb dbs]$

```

now you can create users on this tablespace .

create tablespace in deferent directory

you can stored them in place other that the default path

```
create tablespace [tablespace-name] datafile 'filepath\anyname.dbf' size [sizeyou want]m
```

alter tablespace to increase

you can resize the tablespace if we want using `alter` command to increase the space but be noted to but in consideration the space in physical storage .

as tablespace will reserve the space it configured from disk .

syntax:

```
ALTER DATABASE DATAFILE 'datafile_name' RESIZE size;
```

```
SQL> alter database datafile 'tbs1_data.dbf' resize 10m;
Database altered.
SQL> █
```

```
alter database datafile 'tbs1_data.dbf' resize 10m;
```

When you've already utilized 80MB out of the 100MB available in a tablespace, issuing a RESIZE command to increase it by 10MB will adjust the tablespace size to 90MB, effectively accommodating the additional storage needed. It doesn't expand the tablespace to its original 100MB size; instead, it adapts to your storage requirements.

Conversely, if you apply the RESIZE command to a 100MB tablespace that hasn't stored any data, setting it to 10MB will shrink the tablespace size to match the specified 10MB. In this case, the command alters the tablespace size to exactly what you've designated, as there is no existing data to consider

create user stored in deferent tablespace

now we will create user but we will store the the user in tablespace we have created before .

syntax:

```
CREATE USER [username]
IDENTIFIED BY [password]
DEFAULT TABLESPACE [tablespace_name];
```

```
CREATE USER salah IDENTIFIED BY 101124738 DEFAULT TABLESPACE tbs1;
```

enable autoextend on tablespace

You can create a tablespace that automatically extends its allocated space by a specified value when needed. However, it's crucial to exercise caution when employing this command, as the tablespace will reserve additional space from the disk where it's configured. This means that if not managed prudently, it can consume disk space rapidly, potentially leading to storage issues.

syntax:

```
CREATE TABLESPACE [tablespace_name]
DATAFILE '[datafile_name]' SIZE [initial_size] AUTOEXTEND ON NEXT [next_size];
```

```
CREATE TABLESPACE tbs3 DATAFILE 'tbs3_data.dbf' SIZE 1M AUTOEXTEND ON NEXT 20M;
```

table portioner

if you have tables that you want to store them in tablespace with faster disk space
and store table that is not used a lot in tablespace in slower disk

```
SQL> create tablespace tbs4 datafile '/home/oracle/tbs4_data.dbf' size 200m;
Tablespace created.
SQL> █
```

```
ALTER TABLE table_name MOVE TABLESPACE new_tablespace;
```

how to check the table space size and free space

you can use the below query that will get the space and free space on all tablespace

```
select df.tablespace_name "Tablespace",
totalusedspace "Used MB",
(df.tablespace - tu.totalusedspace) "Free MB",
df.tablespace "Total MB",
round(100 * ( (df.tablespace - tu.totalusedspace) / df.tablespace),2)
"Pct. Free"
from
(select tablespace_name,
round(sum(bytes) / 1048576) TotalSpace
from dba_data_files
group by tablespace_name) df,
(select round(sum(bytes)/(1024*1024)) totalusedspace, tablespace_name
from dba_segments
group by tablespace_name) tu
where df.tablespace_name = tu.tablespace_name;
```

create temporary tablespace

you can create temporary tablespace for temp storing user and tables

```
create tablespace [name] tempfile [filename.dbf] size [size you want]m;
```

```
CREATE TEMPORARY TABLESPACE temporarytablespace
TEMPFILE 'temptbs_file.dbf' SIZE 1000M;
```



```
SQL> create tablespace temporaytabl tempfile 'temptbs_file.dbf' size 1000m;
create tablespace temporaytabl tempfile 'temptbs_file.dbf' size 1000m
*
ERROR at line 1:
ORA-02180: invalid option for CREATE TABLESPACE

SQL> CREATE TEMPORARY TABLESPACE temporarytablespace
TEMPFILE 'temptbs_file.dbf' SIZE 1000M;
3
SQL> █
```

tablespace status

read only tablespace

\$(toc)

this tablespace that only support select command usfull for tables that are not in use or that not used anymore

read write tablespace

this allow read and write you can execute the select and update and insert and delete on the tables on tablespace

offline tablespace

this one that will not allow any action t be done on data on tablespace

drop tablespace

there two way to drop

if you have tablespace that table has constrian or forgenkeys

then you need to use the below command

```
drop tablespace [tablespace - name] including contents and cascade constraints
```

this query doesn't not delete data file it only delete tablespace object

```
DROP TABLESPACE [tablespace_name] INCLUDING CONTENTS AND DATAFILES;
```

```
SQL> DROP TABLESPACE tbs1 INCLUDING CONTENTS AND DATAFILES;

Tablespace dropped.

SQL> SQL> SQL> SQL> █
```

6. Administrating user permission and security

\$(toc)

in oracle the database it self is called users , and users are called users

when create a database it will be created under the user who created the database.

normally in oracle we will create user and then create database under the user this which the called **user database**.

manging user

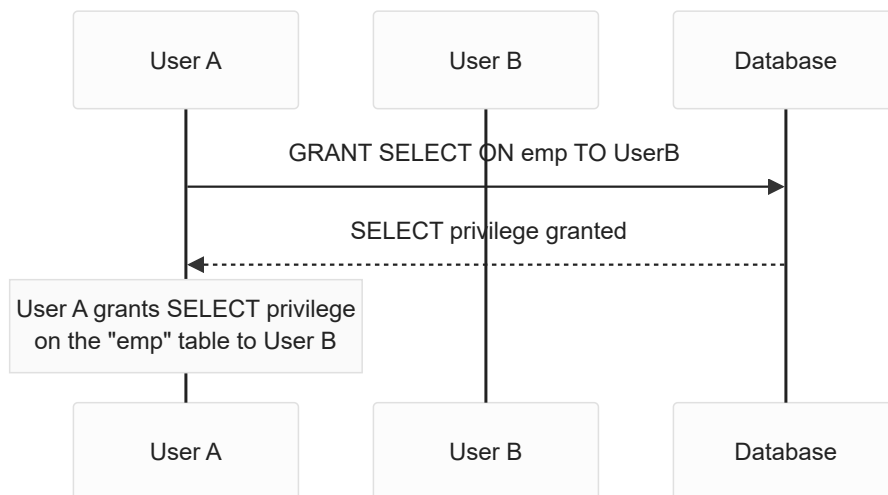
there are two type of user security

below are type of privilege you can grant to the user

1. system privilege

is all the privilege that are granted to user to manage the database engine.

2. object privilege



user privilege on another user , if we have user that has table called emp , then I can give other user select privilege on the user emp table .

In Oracle's SQL syntax, `'GRANT'` and `'REVOKE'` are SQL statements used to control access and permissions for various database objects. These objects can include tables, views, procedures, and more. The `'GRANT'` statement gives specific privileges to a user or role, while the `'REVOKE'` statement takes those privileges away. Here's a more detailed breakdown:

1. GRANT Statement Syntax:

```
GRANT privilege [, privilege, ...]
ON object
TO user [, user, ...]
[WITH ADMIN OPTION];
```

- **'privilege'**: This represents the specific permission you want to grant, such as 'SELECT,' 'INSERT,' 'UPDATE,' 'DELETE,' or 'CREATE.' You can specify multiple privileges separated by commas.

- **'object'**: This refers to the database object (e.g., a table, view, or procedure) on which you want to grant the privileges.
- **'user'**: The user or role to whom you're granting the privileges. You can grant to multiple users or roles simultaneously.
- **'WITH ADMIN OPTION'** (optional): When used, this option allows the recipient to further grant the same privileges to other users or roles. It essentially delegates the privilege management.

2. REVOKE Statement Syntax:

```
REVOKE privilege [, privilege, ...]
ON object
FROM user [, user, ...]
[CASCADE CONSTRAINTS];
```

- **'privilege'**: Similar to 'GRANT,' this specifies the privilege(s) you want to revoke.
- **'object'**: The database object from which you're revoking the privileges.
- **'user'**: The user or role from whom you're revoking the privileges.
- **'CASCADE CONSTRAINTS'** (optional): If you're revoking privileges on a table, this option also removes any dependent constraints owned by the user. Be cautious when using it, as it can have far-reaching consequences.

Here's a practical example. Let's say you want to grant the 'SELECT' and 'INSERT' privileges on the 'employees' table to the user 'Alice' and allow her to further grant these privileges to others:

```
GRANT SELECT, INSERT
ON employees
TO Alice
WITH ADMIN OPTION;
```

This command grants 'Alice' both SELECT and INSERT permissions on the 'employees' table and empowers her to pass on these privileges to additional users if needed.

On the flip side, if you want to remove these privileges from 'Alice,' you'd use the 'REVOKE' statement:

```
REVOKE SELECT, INSERT
ON employees
FROM Alice;
```

This statement takes away the SELECT and INSERT privileges from 'Alice' on the 'employees' table.

Remember, precise management of privileges is crucial for database security and access control. Always use these commands with care and according to your database security policies.

example

create user with admin option

i will create user called mg , the another user called sg

i will grant mg create session privilege and adding with admin option so he can grant other user the same privilege .

what is create session privilege's

This privilege allows a user to establish a connection to the Oracle database. In essence, it's the basic permission needed for a user to log in and interact with the database.

1. create user

```
create user mg identified by mg;
```

2. grant the user create session with admin option

```
grant create session to mg with admin option
```

```
SQL> alter session set "_ORACLE_SCRIPT"=true;

Session altered.

SQL> create user mg identified by mg;

User created.

SQL> grant create session to mg with admin option
      2  ;

Grant succeeded.

SQL> █
```

3. create another user that will we use to test with admin option

```
create user sg identified by sg;
```

4. login to oracle db using mg user

```
[oracle@orcaledb ~]$ sqlplus mg

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 02:13:10 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter password:
Last Successful login time: Mon Jan 15 2024 02:11:18 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> █
```

5. then grant user sg create session using mg user

since we provided the user mg with admin option then the user will be able to grant other user his same privilege .

```
grant create session to sg ;
```

```
[oracle@orcaledb ~]$ sqlplus mg

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 02:13:10 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter password:
Last Successful login time: Mon Jan 15 2024 02:11:18 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> grant create session to sg ;

Grant succeeded.

SQL> █
```

we didn't provide user `sg` with admin privilege's , if we try to grant other user privilege using `sg` then he will not be able to grant the user.

```
SQL> grant create session to mg ;
grant create session to mg
*
ERROR at line 1:
ORA-01031: insufficient privileges

SQL> █
```

revoke permission

now we will revoke the permission using

`revoke` parameter

syntax :

```
revoke [permission] from [user]
```

now will revoke permission [create session] from mg user

```
revoke create session from mg;
```

```
SQL> revoke create session from mg;

Revoke succeeded.

SQL> █
```

now user cannot connect to database

```
SQL> quit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@orcaledb ~]$ sqlplus mg

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 07:01:22 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter password:
ERROR:
ORA-01045: user MG lacks CREATE SESSION privilege; logon denied

Enter user-name: █
```

system privilege

In the realm of Oracle Database, these privileges are akin to the ultimate authority, conferring users with the capability to execute profound administrative operations and wield control over fundamental aspects of the database system. Here's a more in-depth exploration:

CREATE SESSION: This privilege is fundamental , this allow you to connect to database so it must be exist this similar to MSSQL public privilege that allow you to connect to the database .

1. **SYSDBA:** This is the most powerful system privilege, granting full administrative control over the database, including the ability to start and stop the database, perform backup and recovery operations, and modify database parameters.
2. **SYSOPER:** A step below SYSDBA, SYSOPER allows for basic database operations such as starting and shutting down the database. It has fewer administrative capabilities compared to SYSDBA.
3. **CREATE SESSION:** This privilege is fundamental, allowing users to establish a connection and log in to the database. Without it, users can't even access the database.
4. **CREATE TABLE:** Grants the ability to create new tables within a schema. It's often used by application developers to define data structures.
5. **ALTER SYSTEM:** Provides the authority to modify certain global database settings and configuration parameters.
6. **DROP USER:** Permits the removal of user accounts. This privilege is typically held by administrators responsible for user management.
7. **SELECT ANY TABLE:** Allows a user to query any table in the database, regardless of ownership. It's useful in scenarios where users need broad read access.
8. **INSERT, UPDATE, DELETE ANY TABLE:** These privileges enable users to perform data manipulation operations on any table in the database, irrespective of ownership.
9. **EXECUTE ANY PROCEDURE:** Grants the right to execute any stored procedure or function in the database, regardless of the schema in which it resides.

10. **CREATE SESSION, ALTER SESSION:** These privileges pertain to session management and configuration, including the ability to alter one's own session settings.

These system privileges cover a spectrum of actions, from fundamental access (CREATE SESSION) to more advanced administrative and data manipulation capabilities. They are typically assigned based on a user's role and responsibilities within the database environment.

object privilege

here you can give privilege on object of the database to do specific action
such as you can give `select` privilege to user to do select statement on certain table or all tables on object
or update or insert for user to certain table of the database .

example

grant user select statement on table

we will grant select privilege on table called employee to user called mg user

```
grant select on hr.employees to mg
```

```
SQL> grant select on hr.employees to mg

Grant succeeded.

SQL> █
```

```
[oracle@orcaledb ~]$ sqlplus mg

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 08:06:25 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter password:
Last Successful login time: Mon Jan 15 2024 02:13:12 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> select * from hr.employees;

```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY
100	Steven	King					
SKING				1.515.555.0100	17-JUN-13	AD_PRES	24000
				90			

can user share privilege to other user

let's check the user can grant other user grant select on table hr.employees

```
grant select on hr.employees to ahmed
```

```
SQL> grant select on hr.employees to ahmed;  
grant select on hr.employees to ahmed  
*
```

```
ERROR at line 1:  
ORA-01031: insufficient privileges
```

```
SQL> █
```

he cannot because the user don't have with grant option

```
grant select on hr.employees to mg with grant option ;
```

```
SQL> grant select on hr.employees to mg with grant option ;
```

```
Grant succeeded.
```

```
SQL> quit
```

```
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

mg

```
[oracle@orcaledb ~]$ sqlplus mg
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 08:34:07 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Enter password:
```

```
Last Successful login time: Mon Jan 15 2024 08:11:52 -05:00
```

```
Connected to:
```

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> grant select on hr.employees to ahmed;
```

```
Grant succeeded.
```

```
SQL> █
```

revoke privilege from user

now we will revoke select privilege from mg user .


```
REVOKE [privilege] ON [schema].[table] FROM [user];
```

```
revoke select on hr.employees from mg
```

```
[oracle@orcaledb ~]$ sqlplus hr
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Jan 15 08:40:21 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Enter password:
```

```
Last Successful login time: Mon Jan 15 2024 08:32:11 -05:00
```

```
Connected to:
```

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> revoke select on hr.employees from mg;
```

```
Revoke succeeded.
```

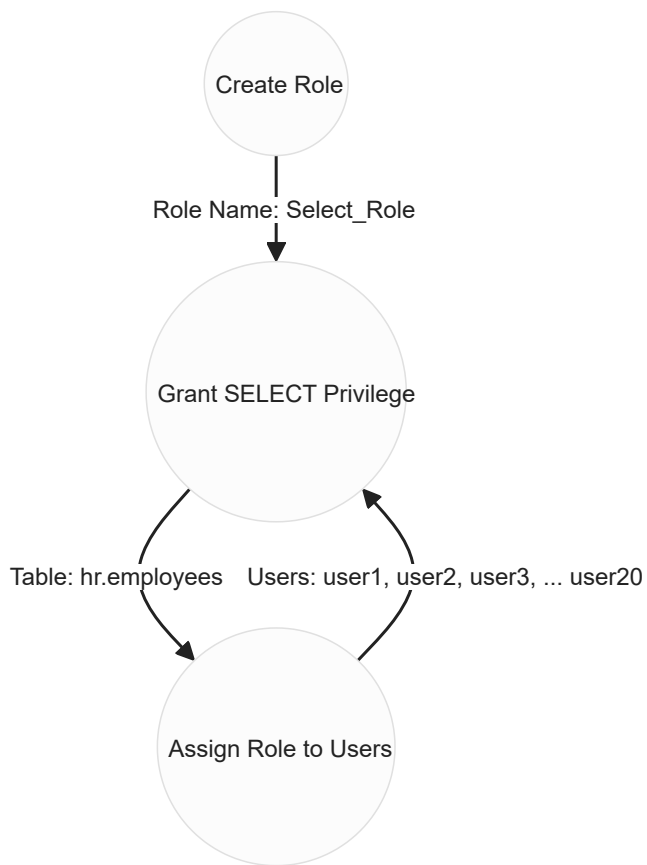
```
SQL> █
```

make role to automate privilege

if you have more than 200 user you want to grant select , this will take time , best option to create role .

role is container containing all privilege you want to assign to specific user , and you can customize this role.

in the role you can assign mutable privilege and assign it to users.



create role

Create a Role: First, create a role named "Select_Role" using SQL:

```
CREATE ROLE Select_Role;
```

```
create role hr_select ;
```

```
SQL> create role hr_select ;  
Role created.  
SQL> █
```

Grant SELECT Privilege: Assign the SELECT privilege on the desired table(s) to the "Select_Role". Let's assume you want to grant SELECT on the employees table in the hr schema:

```
GRANT SELECT ON hr.employees TO Select_Role;
```

```
grant select on hr.employees to hr_select ;
```

```
SQL> grant select on hr.employees to hr_select ;

Grant succeeded.

SQL> █
```

with this you can assignee role to meltable user using the below syntax

```
GRANT Select_Role TO user1, user2, user3, ... user200;
```

```
grant hr_select to mg ;
```

```
SQL> grant hr_select to mg ;

Grant succeeded.

SQL> █
```

predefined roles

Predefined Roles

Role	Privileges Included
CONNECT	CREATE SESSION
RESOURCE	CREATE CLUSTER, CREATE INDEXTYPE, CREATE OPERATOR, CREATE PROCEDURE, CREATE SEQUENCE, CREATE TABLE, CREATE TRIGGER, CREATE TYPE
SCHEDULER_ ADMIN	CREATE ANY JOB, CREATE EXTERNAL JOB, CREATE JOB, EXECUTE ANY CLASS, EXECUTE ANY PROGRAM, MANAGE SCHEDULER
DBA	Most system privileges; several other roles. Do not grant to nonadministrators.
SELECT_ CATALOG_ROLE	No system privileges; HS_ADMIN_ROLE and over 1,700 object privileges on the data dictionary

profile

each user created has preference

Profile	
SESSIONS_PER_USER UNLIMITED	FAILED_LOGIN_ATTEMPTS 5
CPU_PER_SESSION	PASSWORD_LIFE_TIME 60
CPU_PER_CALL	PASSWORD_REUSE_TIME 60
CONNECT_TIME	PASSWORD_REUSE_MAX 5
LOGICAL_READS_PER_SESSION	PASSWORD_VERIFY_FUNCTION <u>verify_function</u>
LOGICAL_READS_PER_CALL	PASSWORD_LOCK_TIME 1/24
PRIVATE_SGA	PASSWORD_GRACE_TIME 10;
COMPOSITE_LIMIT	

session per user:

how many session can single user connect to oracledb.
by default is set as unlimited

cpu-per-session

when user create session it consume cpu
you can edit the amount of cpu the user can consume

cpu-per-call

when you session is open when you make query you will make call to database , here is how many call can user can make.

connection time

how mush time can session can be open if ideal.

fail_login_attempt

how many failed attempt allowed till user is get locked by default is set to 5

8. backup and restore

- [type of backup](#)
 - [1.export regular](#)
 - [2. data pumb](#)
- [how to export and import using datapump](#)
 - [1.create directory and marked it in oracle engin](#)
 - [export schema](#)
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- [import and export normal way](#)
 - [export the normal way](#)
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type of backup

1.Export regular

it produce dump file 'dmp' extensions

allow to export database and save it in any place with out the need to specify the dump directory

2, Data pumb

data pump must specify the directory

it also produce dump file 'dmp' extensions

in this case you have to create folder and then go to database engine and specify the directory

create directory dump as '/path '

once you specify the directory this will be only directory to export and import

and also the user must have privilege

```
sql> (login: Mon Jan 22 02:36:10 2024 from 10.10.10.25)
oracle@oracaledb ~$ ls
tablespace  third data.dpf
oracle@oracaledb ~$ cd /
oracle@oracaledb ~$ mkdir dump
mkdir: cannot create directory 'dump': Permission denied
oracle@oracaledb ~$ mkdir dump
mkdir: cannot create directory 'dump': Permission denied
oracle@oracaledb ~$ ls
oracle@oracaledb ~$ cd /
oracle@oracaledb ~$ ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin share srv sys usr var
oracle@oracaledb ~$ cd /home/
oracle@oracaledb ~$ ls
shared oracle
oracle@oracaledb ~$ cd oracle/
oracle@oracaledb ~$ mkdir dumps
oracle@oracaledb ~$ sqlplus sys/sys as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Jan 27 10:42:18 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> create directory dump as '/home/oracle/dumps/';
Directory created.

SQL>
```

the user `hr` must have dba privilege

```
grant dba to hr
```

How to export and import using datapump

1.Create directory and marked it in oracle engin

first we will create directory for backup.

note that directory must be owned by oracle user .

for this purpose I will create directory in oracle home directory.

```
[oracle@oracledb2 ~]$ cd /
[oracle@oracledb2 /]$ cd home/oracle/
[oracle@oracledb2 ~]$ pwd
/home/oracle
[oracle@oracledb2 ~]$ mkdir dump2
[oracle@oracledb2 ~]$
```

now login to user for this instance i will login using `hr` user that have sample schema and the execute the below command .

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jan 31 01:45:46 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Enter password:
Last Successful login time: Wed Jan 31 2024 01:45:22 -05:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> create directory dumps as '/home/oracle/dump2/';

Directory created.

SQL>
```

```
CREATE DIRECTORY dumps AS '/home/oracle/dump2/';
```

- **CREATE DIRECTORY** : This part of the command initiates the creation of a directory object.
- **dumps** : This is the name of the directory object you are creating. You can choose any name that makes sense for your use case.
- **AS '/home/oracle/dump2/'** : This specifies the actual physical directory path on the server's file system that you want to associate with the directory object. In this example, the directory object named "dumps" is associated with the path '/home/oracle/dump2/' on the server.

syntax :

```
CREATE DIRECTORY directory_name AS 'directory_path';
```

Export schema

to export the schema is `expdp` and can only be done through the terminal .
and you will have to fill the directory which we have submitted in database engine .
which `dumps`
and then fill the filename follow by the file extensions `.dmp`

```
expdp hr/hr@orcl directory=dumps dumpfile=test_exp_jan.dmp
```

```
the command completed successfully
[oracle@orcladb ~]$ expdp hr/hr@prod directory=dumps dumpfile=test_exp_jan.dmp

Export: Release 19.0.0.0.0 - Production on Wed Jan 31 04:02:39 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.

UDE-01017: operation generated ORACLE error 1017
ORA-01017: invalid username/password; logon denied

Username: hr
Password:

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Warning: Oracle Data Pump operations are not typically needed when connected to the root or seed of a container database.

Starting "HR"."SYS_EXPORT_SCHEMA_01": hr/***** directory=dumps dumpfile=test_exp_jan.dmp
Processing object type SCHEMA_EXPORT/TABLE/TABLE DATA
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/SYSTEM GRANT
Processing object type SCHEMA_EXPORT/ROLE GRANT
Processing object type SCHEMA_EXPORT/DEFAULT ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
```

syntax

```
expdp [username[/password]@connect_identifier] DIRECTORY=directory_object DUMPFILE=dumpfile_name
```

- `username[/password]` : The Oracle username and optional password for the database connection.
- `@connect_identifier` : The connection identifier for the Oracle database.
- `DIRECTORY=directory_object` : Specifies the Oracle directory object representing the directory where the export file will be placed.
- `DUMPFILE=dumpfile_name` : Indicates the name of the dump file to be generated.

Remember to replace username, password, connect_identifier, directory_object, and dumpfile_name with your specific database and export file details.

the

```

Password:
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Warning: Oracle Data Pump operations are not typically needed when connected to the root or seed of a container database.

Starting "HR"."SYS_EXPORT_SCHEMA_01": hr/***** directory=dumps dumpfile=test_exp_jan.dmp
Processing object type SCHEMA_EXPORT/TABLE/TABLE DATA
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/SYSTEM GRANT
Processing object type SCHEMA_EXPORT/ROLE GRANT
Processing object type SCHEMA_EXPORT/DEFAULT ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROACT_SCHEMA
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Master table "HR"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
*****
Dump file set for HR.SYS_EXPORT_SCHEMA_01 is:
/home/oracle/dump2/test_exp_jan.dmp
Job "HR"."SYS_EXPORT_SCHEMA_01" successfully completed at Wed Jan 31 04:04:08 2024 elapsed 0 00:01:20

[oracle@orcaledb ~]$ cd /home/oracle/dump2/
[oracle@orcaledb dump2]$ ls
export.log test_exp_jan.dmp
[oracle@orcaledb dump2]$

```

```

[oracle@orcaledb ~]$ cd /home/oracle/dump2/
[oracle@orcaledb dump2]$ ls
export.log test_exp_jan.dmp
[oracle@orcaledb dump2]$

```

What does the export contain

the export will contain the table , roles , views , database links ,index.

you can see the details during the export processes.

```

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Warning: Oracle Data Pump operations are not typically needed when connected to the root or seed of a container database.

Starting "HR"."SYS_EXPORT_SCHEMA_01": hr/***** directory=dumps dumpfile=test_exp_jan.dmp
Processing object type SCHEMA_EXPORT/TABLE/TABLE DATA
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/SYSTEM GRANT
Processing object type SCHEMA_EXPORT/ROLE GRANT
Processing object type SCHEMA_EXPORT/DEFAULT ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROACT_SCHEMA
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Master table "HR"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
*****
Dump file set for HR.SYS_EXPORT_SCHEMA_01 is:
/home/oracle/dump2/test_exp_jan.dmp
Job "HR"."SYS_EXPORT_SCHEMA_01" successfully completed at Wed Jan 31 04:04:08 2024 elapsed 0 00:01:20

```

Import dump file

as we did before , we exported everting under user hr .

now we want to export, to achieve that we will need to import the dump under another user
for this purpose i will create user and grant it the necessary privilege .

then I will import `hr` user dump to that user .

```
SQL> alter user ahmed identified by ahmed;

User altered.

SQL> grant dbat to ahmed;
grant dbat to ahmed
      *
ERROR at line 1:
ORA-01917: user or role 'DBAT' does not exist

SQL> grant dba to ahmed;

Grant succeeded.

SQL> grant create session to ahmed;

Grant succeeded.

SQL> exit
```

since I have use `expdp` to export files ,

then we will use `impdp`

importing for dump created using `expdp` can only be imported using `impdp` .

example :

```
impdp ahmed/ahmed@prod directory=dumps dumpfile=test_exp_jan.dmp remap_schema=hr:ahmed
logfile=import.log
```

- **username/password:** The Oracle username and password used to connect to the target database.
- **@connect_identifier:** The connection identifier for the Oracle database.

-DIRECTORY=directory_object: Specifies the Oracle directory object representing the directory from which the export file will be read during import.

- **DUMPFILE=dumpfile_name:** Indicates the full path to the dump file to be used for the import operation.

-REMAP_SCHEMA=source_schema:target_schema: Allows you to map the source schema in the dump file to the target schema in the database during the import process

so for our case we exported from `hr` user and planning to import `ahmed` user

```

Import: Release 19.0.0.0.0 - Production on Wed Jan 31 09:29:03 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.

UDI-01017: operation generated ORACLE error 1017
ORA-01017: invalid username/password; logon denied

Username: ahmed
UDI-00002: invalid username or password

Username: ahmed
Password:

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Warning: Oracle Data Pump operations are not typically needed when connected to the root or seed of a container database.

Master table "AHMED"."SYS_IMPORT_FULL_01" successfully loaded/unloaded
Starting "AHMED"."SYS_IMPORT_FULL_01": ahmed/***** directory=dumps dumpfile=test_exp_jan.dmp remap_schema=hr:ahmed
Processing object type SCHEMA_EXPORT/SYSTEM GRANT
Processing object type SCHEMA_EXPORT/ROLE GRANT
Processing object type SCHEMA_EXPORT/DEFAULT ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Job "AHMED"."SYS_IMPORT_FULL_01" successfully completed at Wed Jan 31 09:29:18 2024 elapsed 0 00:00:07

[oracle@oracledb dump2]$ █

```

Export dump with password

if you have very critical database with sensitive data that you don't want to be seen by unauthorised people

syntax:

```

expdp hr/hr@prod schemas=hr directory=dumps dumpfile=[file].dmp logfile=[name].log encryption=all
encryption_password=[password]

```

-expdp: This is the command used to invoke Oracle Data Pump, a utility for exporting and importing data in Oracle databases.

-hr/hr@prod: This section specifies the username and password used to connect to the Oracle database. It's in the format username/password@database. In this case, it's connecting as user hr with the password hr to the prod database.

-schemas=hr: This option specifies the schema (hr) from which data will be exported. A schema is a logical container for database objects within the Oracle database.

-directory=dumps: Here, you specify the Oracle directory named dumps. This directory should have been previously created in the database and is used as the location where the Data Pump utility will store the export files.

-dumpfile=[file].dmp: This part defines the name of the dump file that will be generated during the export. You should replace [file] with your desired file name. The .dmp extension is standard for Oracle Data Pump export files.

-logfile=[name].log: Similar to the dumpfile, this option specifies the name of the log file that will be generated during the export. Replace [name] with your preferred log file name. The .log extension is commonly used for log files.

-encryption=all: This indicates that data in the export file will be encrypted. The encryption=all option ensures that all exported data is encrypted for security purposes.

-encryption_password=[password]: Here, you need to specify the encryption password. Make sure to replace [password] with the actual encryption password you intend to use. This password will be required to decrypt the exported data when needed.

Import and export normal way

export the normal way

In here you don't need to inform the database engine what is the directory for storing the backup in here you can specify the path directory in export command

syntax:

```
exp username/password file=[filename].dmp log=[file-name].log
```

example:

```
exp hr/hr file=/home/oracle/dump2/backup2.dmp log=/home/oracle/dump2/backup2.log
```

```
[oracle@orcaledb ~]$ ls
dump2  dumps  tablespace  tbs4_data.dbf
[oracle@orcaledb ~]$ exp hr/hr file=/home/oracle/dump2/backup2.dmp log=/home/oracle/dump2/backup2.log
```

```
Export: Release 19.0.0.0.0 - Production on Tue Feb 6 08:17:21 2024
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.
```

```
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
Export done in US7ASCII character set and AL16UTF16 NCHAR character set
server uses AL32UTF8 character set (possible charset conversion)
```

```
About to export specified users ...
. exporting pre-schema procedural objects and actions
. exporting foreign function library names for user HR
. exporting PUBLIC type synonyms
. exporting private type synonyms
. exporting object type definitions for user HR
About to export HR's objects ...
. exporting database links
. exporting sequence numbers
. exporting cluster definitions
. about to export HR's tables via Conventional Path ...
```

```
EXP-00091: Exporting questionable statistics.
EXP-00091: Exporting questionable statistics.
EXP-00091: Exporting questionable statistics.
EXP-00091: Exporting questionable statistics.
. . exporting table          REGIONS          5 rows exported
EXP-00091: Exporting questionable statistics.
EXP-00091: Exporting questionable statistics.
. exporting synonyms
. exporting views
. exporting stored procedures
. exporting operators
. exporting referential integrity constraints
. exporting triggers
. exporting indextypes
. exporting bitmap, functional and extensible indexes
. exporting partitables actions
. exporting materialized views
. exporting snapshot logs
. exporting job queues
. exporting refresh groups and children
. exporting dimensions
. exporting post-schema procedural objects and actions
. exporting statistics
Export terminated successfully with warnings.
[oracle@orcaledb ~]$ cd /home/oracle/dump2/
[oracle@orcaledb dump2]$ ls
backup2.dmp  backup2.log  export.log  import.log  test_exp_jan.dmp
[oracle@orcaledb dump2]$
```

Import the normal way

The below is syntax for importing using the normal exportnote that the import only work with export execute with exp command
Syntax

```
imp userid=username/password full=y file='path of dmp export file '
```

Examble:

```
imp userid=ahmed/ahmed full=y file='/home/oracle/dump2/backup2.dmp'
```

```
[oracle@orcaledb dump2]$ imp userid=ahmed/ahmed full=y file='/home/oracle/dump2/backup2.dmp'
Import: Release 19.0.0.0.0 - Production on Wed Feb 7 03:11:07 2024
Version 19.3.0.0.0

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Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

Export file created by EXPORT:V19.00.00 via conventional path

Warning: the objects were exported by HR, not by you

import done in US7ASCII character set and AL16UTF16 NCHAR character set
import server uses AL32UTF8 character set (possible charset conversion)
IMP-00403:

Warning: This import generated a separate SQL file "import_sys" which contains DDL that failed due to a privilege issue.

. importing HR's objects into AHMED
. . importing table          "COUNTRIES"          25 rows imported
. . importing table          "DEPARTMENTS"         27 rows imported
. . importing table          "EMPLOYEES"          107 rows imported
. . importing table          "JOBS"                19 rows imported
```

```
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. . importing table          "DEPARTMENTS"         27 rows imported
. . importing table          "EMPLOYEES"          107 rows imported
. . importing table          "JOBS"                19 rows imported
. . importing table          "JOB_HISTORY"          10 rows imported
. . importing table          "LOCATIONS"           23 rows imported
. . importing table          "REGIONS"              5 rows imported
About to enable constraints
Import terminated successfully with warnings.
[oracle@orcaledb dump2]$
```